

**BASIC
MAINTENANCE
INFORMATION**



Standby and Prime Power Generators



MODELS:
LIQUID-COOLED UNITS

STANDBY GENERATORS

SAFETY

Throughout this publication, DANGER, WARNING, and CAUTION blocks are used to alert the mechanic to special instructions concerning a particular service or operation that might be hazardous if performed incorrectly or carelessly. Observe them carefully. Their definitions are as follows:

DANGER!

After this heading, read instructions that, if not strictly complied with, will result in serious personal injury, including death.

WARNING!

After this heading, read instructions that, if not strictly complied with, could result in serious personal injury, including death.

CAUTION!

After this heading, read instructions that, if not strictly complied with, might result in minor or moderate injury.

Four commonly used safety symbols accompany the DANGER, WARNING and CAUTION blocks. The type of information each indicates follows:

 **This symbol points out important safety information that, if not followed, could endanger personal safety and/or property of others.**

 **This symbol points out potential explosion hazard.**

 This symbol points out potential fire hazard.

 **This symbol points out potential electrical shock hazard.**

These “Safety Alerts” alone cannot eliminate the hazards that they signal. Strict compliance with these special instructions plus “common sense” are major accident prevention measures.

READ THIS MANUAL THOROUGHLY

This SERVICE MANUAL has been written and published by Generac to aid our dealers' mechanics and company service personnel when servicing the products described herein.

It is assumed that these personnel are familiar with the servicing procedures for these products, or like or similar products manufactured and marketed by Generac. That they have been trained in the recommended servicing procedures for these products, including the use of common hand tools and any special Generac tools or tools from other suppliers.

Generac could not possibly know of and advise the service trade of all conceivable procedures by which a service might be performed and of the possible hazards and/or results of each method. We have not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or tool not recommended by Generac must first satisfy themselves that neither his nor the products safety will be endangered by the service procedure selected.

All information, illustrations and specifications in this manual are based on the latest product information available at the time of publication.

When working on these products, remember that the electrical system and engine ignition system are capable of violent and damaging short circuits or severe electrical shocks. If you intend to perform work where electrical terminals could be grounded or touched, the battery cables should be disconnected at the battery.

Any time the intake or exhaust openings of the engine are exposed during service, they should be covered to prevent accidental entry of foreign material. Entry of such materials will result in extensive damage when the engine is started.

During any maintenance procedure, replacement fasteners must have the same measurements and strength as the fasteners that were removed. Metric bolts and nuts have numbers that indicate their strength. Customary bolts use radial lines to indicate strength while most customary nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction and possible injury.

NOTE: Special NOTES appear in bold type throughout this publication. While not pertaining to safety, they emphasize procedures, circumstances or specifications that require special attention.

REPLACEMENT PARTS

When servicing this equipment, it is extremely important that all components be properly installed and tightened. If improperly installed and tightened, sparks could ignite fuel vapors from fuel system leaks.

TABLE OF CONTENTS

<i>Foreword</i>	<i>Inside Front Cover</i>
<i>Safety Rules</i>	<i>2</i>
<i>Service Schedule</i>	<i>5</i>
<i>Visual Inspection</i>	<i>9</i>
<i>Oil Measurement Procedure</i>	<i>9</i>
<i>Battery Handling and Maintenance</i>	<i>10</i>
<i>Belts</i>	<i>13</i>
<i>Filters</i>	<i>17</i>
<i>Spark Plugs</i>	<i>24</i>
<i>Hoses</i>	<i>32</i>
<i>Coolant</i>	<i>35</i>



Important Safety Notice

Proper service and repair is important to the safe, economical and reliable operation of all standby electric power systems. The troubleshooting, testing and servicing procedures recommended by Generac and described in this manual are effective methods of performing such operations. Some of these operations or procedures may require the use of specialized equipment. Such equipment should be used when and as recommended.

It is important to note that this manual contains various DANGER, CAUTION, and NOTE blocks. These should be read carefully in order to minimize the risk of personal injury or to prevent improper methods or practices from being used. Use of improper or unauthorized practices may damage equipment or render it unsafe. The DANGER, CAUTION and NOTE blocks are not exhaustive. Generac could not possibly know, evaluate and advise the service trade of all conceivable ways in which operations described in this manual might be accomplished or of the possible hazardous consequences of each way. Consequently, Generac has not taken any such broad evaluation. Accordingly, anyone who uses any troubleshooting, testing or service procedure that is not recommended by Generac must first satisfy himself that neither his safety nor the equipment's safety will be jeopardized by the procedure or the method he selects.



SAVE THESE INSTRUCTIONS – *The manufacturer suggests that these rules for safe operation be copied and posted in potential hazard areas. Safety should be stressed to all operators and potential operators of this equipment.*

Study these SAFETY RULES carefully before installing, operating or servicing this equipment. Become familiar with the *Owner's Manual* and with the unit. The generator can operate safely, efficiently and reliably only if it is properly installed, operated and maintained. Many accidents are caused by failing to follow simple and fundamental rules or precautions.

Generac cannot anticipate every possible circumstance that might involve a hazard. The warnings in this manual, and on tags and decals affixed to the unit are, therefore, not all-inclusive. If using a procedure, work method or operating technique that Generac does not specifically recommend, ensure that it is safe for others. Also make sure the procedure, work method or operating technique utilized does not render the generator unsafe.

— **DANGER** —



Despite the safe design of this generator, operating this equipment imprudently, neglecting its maintenance or being careless can cause possible injury or death. Permit only responsible and capable persons to install, operate or maintain this equipment.



Potentially lethal voltages are generated by these machines. Ensure all steps are taken to render the machine safe before attempting to work on the generator.



Parts of the generator are rotating and/or hot during operation. Exercise care near running generators.



GENERAL HAZARDS

- For safety reasons, Generac recommends that this equipment be installed, serviced and repaired by an authorized service dealer or other competent, qualified electrician or installation technician who is familiar with applicable codes, standards and regulations. The operator also must comply with all such codes, standards and regulations.
- Installation, operation, servicing and repair of this (and related) equipment must always comply with applicable codes, standards, laws and regulations. Adhere strictly to local, state and national electrical and building codes. Comply with regulations the Occupational Safety and Health Administration (OSHA) has established. Also, ensure that the generator is installed, operated and serviced in accordance with the manufacturer's instructions and recommendations. Following installation, do nothing that might render the unit unsafe or in noncompliance with the aforementioned codes, standards, laws and regulations.
- The engine exhaust fumes contain carbon monoxide gas, which can be DEADLY. This dangerous gas, if breathed in sufficient concentrations, can cause unconsciousness or even death. For that reason, adequate ventilation must be provided. Exhaust gases must be piped safely away from any building or enclosure that houses the generator to an area where people, animals, etc., will not be harmed. This exhaust system must be installed properly, in strict compliance with applicable codes and standards.
- Keep hands, feet, clothing, etc., away from drive belts, fans, and other moving or hot parts. Never remove any drive belt or fan guard while the unit is operating.
- Adequate, unobstructed flow of cooling and ventilating air is critical to prevent buildup of explosive gases and to ensure correct generator operation. Do not alter the installation or permit even partial blockage of ventilation provisions, as this can seriously affect safe operation of the generator.
- Keep the area around the generator clean and uncluttered. Remove any materials that could become hazardous.
- When working on this equipment, remain alert at all times. Never work on the equipment when physically or mentally fatigued.
- Inspect the generator regularly, and promptly repair or replace all worn, damaged or defective parts using only factory-approved parts.
- Before performing any maintenance on the generator, disconnect its battery cables to prevent accidental start-up. Disconnect the cable from the battery post indicated by a NEGATIVE, NEG or (–) first. Reconnect that cable last.
- Never use the generator or any of its parts as a step. Stepping on the unit can stress and break parts, and may result in dangerous operating conditions from leaking exhaust gases, fuel leakage, oil leakage, etc.

ELECTRICAL HAZARDS

- All generators covered by this manual produce dangerous electrical voltages and can cause fatal electrical shock. Utility power delivers extremely high and dangerous voltages to the transfer switch as well as the standby generator. Avoid contact with bare wires, terminals, connections, etc., on the generator as well as the transfer switch, if applicable. Ensure all appropriate covers, guards and barriers are in place before operating the generator. If work must be done around an operating unit, stand on an insulated, dry surface to reduce shock hazard.
- Do not handle any kind of electrical device while standing in water, while barefoot, or while hands or feet are wet. **DANGEROUS ELECTRICAL SHOCK MAY RESULT.**
- If people must stand on metal or concrete while installing, operating, servicing, adjusting or repairing this equipment, place insulative mats over a dry wooden platform. Work on the equipment only while standing on such insulative mats.
- The National Electrical Code (NEC), Article 250 requires the frame and external electrically conductive parts of the generator to be connected to an approved earth ground and/or grounding rods. This grounding will help prevent dangerous electrical shock that might be caused by a ground fault condition in the generator set or by static electricity. Never disconnect the ground wire.



- Wire gauge sizes of electrical wiring, cables and cord sets must be adequate to handle the maximum electrical current (ampacity) to which they will be subjected.
- Before installing or servicing this (and related) equipment, make sure that all power voltage supplies are positively turned off at their source. Failure to do so will result in hazardous and possibly fatal electrical shock.
- Connecting this unit to an electrical system normally supplied by an electric utility shall be by means of a transfer switch so as to isolate the generator electric system from the electric utility distribution system when the generator is operating. Failure to isolate the two electric system power sources from each other by such means will result in damage to the generator and may also result in injury or death to utility power workers due to backfeed of electrical energy.
- Generators installed with an automatic transfer switch will crank and start automatically when NORMAL (UTILITY) source voltage is removed or is below an acceptable preset level. To prevent such automatic start-up and possible injury to personnel, disable the generator's automatic start circuit (battery cables, etc.) before working on or around the unit. Then, place a "Do Not Operate" tag on the generator control panel and on the transfer switch.
- In case of accident caused by electric shock, immediately shut down the source of electrical power. If this is not possible, attempt to free the victim from the live conductor. **AVOID DIRECT CONTACT WITH THE VICTIM.** Use a nonconducting implement, such as a dry rope or board, to free the victim from the live conductor. If the victim is unconscious, apply first aid and get immediate medical help.
- Never wear jewelry when working on this equipment. Jewelry can conduct electricity resulting in electric shock, or may get caught in moving components causing injury.



FIRE HAZARDS



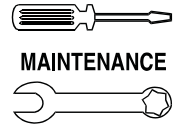
- Keep a fire extinguisher near the generator at all times. Do NOT use any carbon tetra-chloride type extinguisher. Its fumes are toxic, and the liquid can deteriorate wiring insulation. Keep the extinguisher properly charged and be familiar with its use. If there are any questions pertaining to fire extinguishers, consult the local fire department.



EXPLOSION HAZARDS



- Properly ventilate any room or building housing the generator to prevent build-up of explosive gas.
- Do not smoke around the generator. Wipe up any fuel or oil spills immediately. Ensure that no combustible materials are left in the generator compartment, or on or near the generator, as FIRE or EXPLOSION may result. Keep the area surrounding the generator clean and free from debris.
- Generac generator sets may operate using one of several types of fuels. All fuel types are potentially FLAMMABLE and/or EXPLOSIVE and should be handled with care. Comply with all laws regulating the storage and handling of fuels. Inspect the unit's fuel system frequently and correct any leaks immediately. Fuel supply lines must be properly installed, purged and leak tested according to applicable fuel-gas codes before placing this equipment into service.
- Diesel fuels are highly FLAMMABLE. Gaseous fluids such as natural gas and liquid propane (LP) gas are extremely EXPLOSIVE. Natural gas is lighter than air, and LP gas is heavier than air; install leak detectors accordingly.



SERVICE SCHEDULE

22 KW - 150 KW GASEOUS STATIONARY EMERGENCY GENERATOR

The following is a recommended maintenance schedule for Gaseous Stationary Emergency Generator sets from 22kW to 150 kW in size. The established intervals in the schedule are the maximum recommended when the unit is used in an average service application. They will need to be decreased (performed more frequently) if the unit is used in a severe application. Use calendar time, from the previous maintenance interval to determine the next required maintenance interval.

Service Maintenance Interval Information:

The various service maintenance intervals are designated by interval numbers as follows:

- 1 An early inspection of the generator set to insure it is ready to operate when required and to identify any potential problem areas.



This inspection may be performed by the end user providing the following safety steps are taken to prevent the engine from starting automatically without warning:

To prevent injury, perform the following steps in the order indicated before starting any maintenance:

- Disable the generator set from starting and/or connecting to the load by setting the control panel Auto/Off/Manual switch to the "OFF" position.
- Remove the 15 amp control panel fuse.
- Turn off the battery charger.*
- Remove the negative battery cable.
- Open circuit breaker



**** The battery charger must be turned off BEFORE removing the battery cable to prevent an over current condition from burning out sensitive control panel components and circuits.***

Following all maintenance, reverse these steps to insure the unit is returned to standby setup for normal operation when required.

- 2 A wear-in service inspection of the generator set to insure it is ready to operate and carry the load when required, and to identify any potential problem areas.

*Performed **ONLY ONCE** following the first three months or the first 30 hours of operation after purchase of the unit.*

This inspection contains some maintenance tasks which require special tools, equipment, and/or knowledge to accomplish and should be performed only by a Service Dealer.

- 3 An operational inspection of the generator set to insure it is ready to operate and carry the load when required, and to identify any potential problem areas.

Performed semi-annually or following each 50 hours of operation of the unit.

This inspection contains some maintenance tasks which require special tools, equipment, and/or knowledge to accomplish and should be performed only by a Service Dealer.

- 4 A mid-level inspection of the generator set to insure it is ready to operate and carry the load when required, and to identify any potential problem areas.

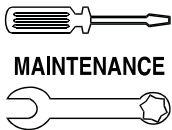
Performed annually or following each 100 hours of operation of the unit.

This inspection contains some maintenance tasks which require special tools, equipment, and/or knowledge to accomplish and should be performed only by a Service Dealer.

- 5 A comprehensive inspection of the generator set to insure it is properly serviced and ready to operate and carry the load when required, and to identify any potential problem areas.

Performed annually or following each 250 hours of operation of the unit.

This inspection contains some maintenance tasks which require special tools, equipment, and/or knowledge to accomplish and should be performed only by a Service Dealer.



MAINTENANCE

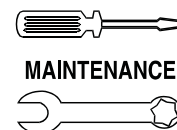
BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS

Generac Power Systems

Maintenance Tasks	Level 1		Level 2		Level 3		Level 4		Level 5	
	Recommended to be done monthly/ 10 hrs.	Task Comp. (Date-Initials)	Required to be done 3 months/ Break-in 30 hrs.	Task Comp. (Date-Initials)	Required to be done Semi-annually/ 50 hrs.	Task Comp. (Date-Initials)	Required to be done Annually/ 100 hrs.	Task Comp. (Date-Initials)	Required to be done Bi-annually/ 250 hrs.	Task Comp. (Date-Initials)
1. Disable the unit from operating per the first page warning.										
2. Check the engine oil level. Adjust as necessary*.										
3. Check the engine coolant level. Adjust as necessary.										
4. Check the engine coolant thermal protection level. Correct as necessary.										
5. Check the natural gas delivery system for leaks and correct pressure on gas engine driven units. Tighten connections as necessary.										
6. Check the air inlets and outlets of the enclosure and radiator for debris. Clean as necessary.										
7. Check the battery electrolyte level and specific gravity if accessible. Adjust as necessary.										
8. Check the battery posts, cables, and charger for loose connections, corrosion, and proper operation. Correct as necessary.										
9. Check the unit wiring for loose connections, corrosion, and damage. Correct as necessary.										

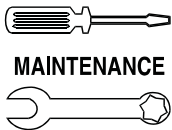
*See procedure to accurately measure oil level.

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS
Generac Power Systems



Maintenance Tasks	Level 1		Level 2		Level 3		Level 4		Level 5	
	Recom- mended to be done monthly/ 10 hrs.	Task Comp. (Date- Initials)	Required to be done 3 months/ Break-in 30 hrs.	Task Comp. (Date- Initials)	Required to be done Semi- annually/ 50 hrs.	Task Comp. (Date- Initials)	Required to be done Annually/ 100 hrs.	Task Comp. (Date- Initials)	Required to be done Bi- annually/ 250 hrs.	Task Comp. (Date- Initials)
10. Check the engine accessory drive belts and fan coupling device if equipped for correct tension, wear, weather cracking, and damage. Replace as necessary.										
11. Check the engine valve clearance/lash. Adjust as necessary.**										
12. Visually inspect the unit looking for leaks, wear or damage, loose connections or components, and corrosion. Correct as necessary.										
13. Test the engine and transfer switch safety devices. Correct and/or adjust as necessary.										
14. Initiate an automatic start and transfer of the unit to site load and exercise it for at least 1 hour looking for leaks, loose connections or components, and abnormal operating conditions. Correct as necessary.										
15. Replace the engine accessory drive belts.										
16. Check gearbox oil level (if equipped).										
17. Change gearbox oil (if equipped).										

** Not required for engines equipped with hydraulic lifters. See the "Specification" section for lifter type.

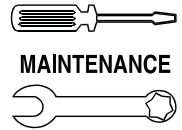


MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS Generac Power Systems

Maintenance Tasks	Level 1 Recom- mended to be done monthly/ 10 hrs.	Task Comp. (Date- Initials)	Level 2 Required to be done 3 months/ Break-in 30 hrs.	Task Comp. (Date- Initials)	Level 3 Required to be done Semi- annually/ 50 hrs.	Task Comp. (Date- Initials)	Level 4 Required to be done Annually/ 100 hrs.	Task Comp. (Date- Initials)	Level5 Required to be done Bi- annually/ 250 hrs.	Task Comp. (Date- Initials)
18. Start and exercise the unit at full rated load (use a load bank if the site load is not enough) for at least 2 hours looking for leaks, loose connections or components, and abnormal operating conditions. Correct as necessary.										
19. Perform an engine oil analysis (send a sample to a lab for results). Change the engine oil and filters if the analysis results indicate this is required.										
20. Change the engine oil.										
21. Replace the engine oil filter(s).										
22. Replace engine spark plugs. Clean and re-gap or replace as necessary.										
23. Replace the engine air filter(s).										
24. Perform a 5 minute no-load operational run of the unit looking for any post service problems.										
25. Return the unit to standby setup for operation when required.										

*Break in only.



VISUAL INSPECTION

When first arriving at a site, look at the condition of the area around the generator. If there is trash/debris, weeds or long grass, it should be removed/cut.

When you open the doors and access panels of the generator use a flashlight to visually inspect inside the compartment. Look for trash, debris, nests or nesting indications. Any foreign objects inside the generator enclosure need to be removed.

Using a flashlight, visually inspect the engine and radiator looking for signs of fluid leakage. Inspect all hose connections, tube connections and engine component seams. Dirt clings to oil and coolant traces so provides a way to trace leaks. Inspect the bottom of the generator enclosure for signs of leakage.

If you find indications of leakage trace it to its source. Occasionally hose clamps and other fluid fittings/couplings need to be tightened. If there are signs of major leakage you will need to determine why the large leak occurred and repair it. If the leak is occurring at an engine component seam, try to establish how severe the leak is. Some gasket materials allow a minor amount of leakage when first compressed. Clean any traces of the leakage and make notes about where it was found. During the next inspection, look for leak traces in those areas. If there is a continuing leak, or if the leakage increases a solution may have to be found. Discuss this with the manufacturer or servicing dealer.

OIL LEVEL MEASUREMENT PROCEDURE

HOW TO ACCURATELY CHECK OIL

1. After running the engine, remove the dipstick and let the engine sit for at least 10 minutes to allow oil to drain.
2. Wipe and insert the dipstick, wait at least 10 seconds for oil to stabilize, remove the dipstick and read the "top" flat surface (rather than the underside flat surface).

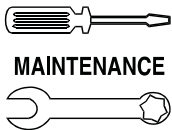
When adding oil, allow at least 10 minutes for oil to drain to the pan.

When changing oil

1. Refill the unit with the amount of oil specified.
2. Start and run the engine for about one minute.
3. Turn off the engine and let it sit for at least 10 minutes to allow oil to drain to pan.
4. Remove the dipstick and read the "top" flat surface (rather than the underside flat surface).
5. Adjust level as needed.

WAYS TO GET ERRONEOUS OIL READINGS

1. Using the underside of the dipstick (read the top side of the dipstick).
2. Remove the dipstick, quickly re-insert it and immediately remove it for a reading (wait 10 minutes before re-inserting).
3. Read dipstick right after shutting off engine (wait at least 10 minutes).
4. Not fully inserting dipstick (make sure dipstick is inserted completely into dipstick tube).



MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS

Generac Power Systems

BATTERY HANDLING AND MAINTENANCE

It is important that all labeling on the battery is carefully read, understood and complied with. The format of the following labels and symbols is common to most brands of lead acid batteries (see Figure 1).



Lead-acid batteries contain a sulfuric acid electrolyte, which is a highly corrosive poison and will produce gas when recharged and explode if ignited. When working with batteries, you need to wear glasses, have plenty of ventilation, remove your jewelry, and exercise caution. Do NOT allow battery electrolyte to mix with salt water. Even small quantities of this combination will produce chlorine gas that can KILL you! Please follow the manufacturer's instructions for testing, installing, discharging, charging, equalizing and maintaining batteries.

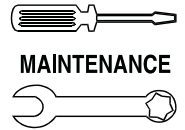
- Performing preventive maintenance on batteries is easy and should occur once a month. Here are some simple steps to maintain your battery:
- The plates need to be covered at all times to prevent sulfation and reduce the possibility of an internal battery explosion.

For non-sealed wet batteries (with filler caps), if the electrolyte levels are low, allow the battery to cool to room temperature first and then add only distilled, deionized or demineralized water to the level indicated by the battery manufacturer or to within 1/4 to 3/8 inch (6 to 10 mm) below the bottom of the filler tubes (vent wells or splash barrels). Avoid overfilling, especially in hot weather, because the heat will cause the electrolyte to expand and overflow.

Figure 1 — Battery Labelling



	Explosive gases		Read relevant instructions
	Eye protection must be WORN.		Keep away from children
	No smoking or naked flames.		Do not dispose of as household waste.
	Corrosive acid		Recycle (via recognized disposal system).
	Flush eyes immediately when contacted with acid		Electrical current may cause injury to personnel
	Caution/important notice.		



NOTE:

In an emergency, use rain water rather than residential reverse osmosis (RO) from residential systems or tap water because rain water does not contain calcium or magnesium. Using RO or tap water to refill batteries can produce calcium sulfate crystals that can fill the pores and coat the plates. State-of-Charge (SoC) readings will be inaccurate immediately after the addition of water, recharges or discharges.

- Tighten loose hold-down clamps, battery terminals and connectors.
- Remove any corrosion, lead oxidation, paint or rust with a brass wire battery brush (brushing the corrosion away from you) or a "ScotchBrite" pad from the terminal's mating surfaces on both ends of each of the battery cables, battery posts or terminals, and engine grounding strap connections. (A stiff steel wire brush may damage protective lead plating on copper connectors or terminals.) Heavy corrosion can be neutralized with a mixture of one pound of baking soda (bicarbonate of soda) to one gallon of warm water. Bare metal to metal mating surfaces are required for good current conductivity.

Figure 2 — Clean Battery Terminals

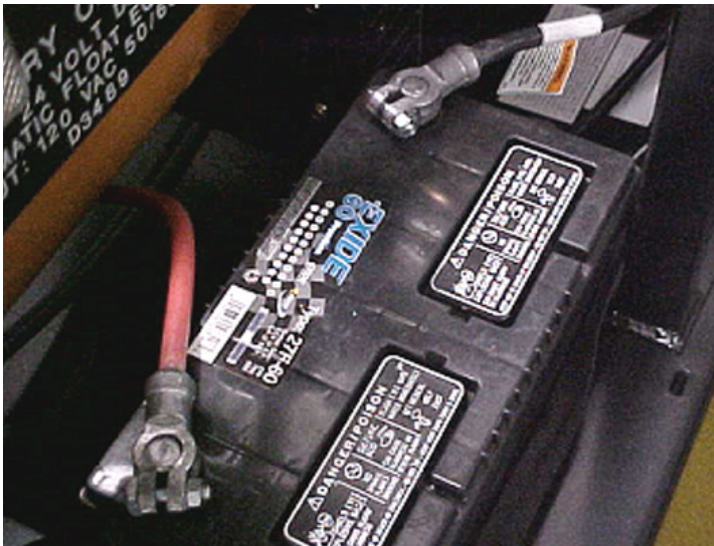


Figure 3 — Corroded Terminal

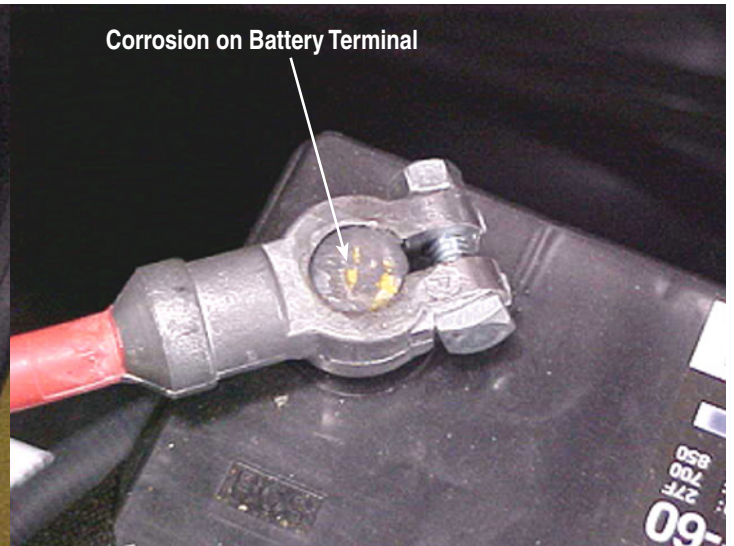


Figure 4 — Corrosion Under Terminal

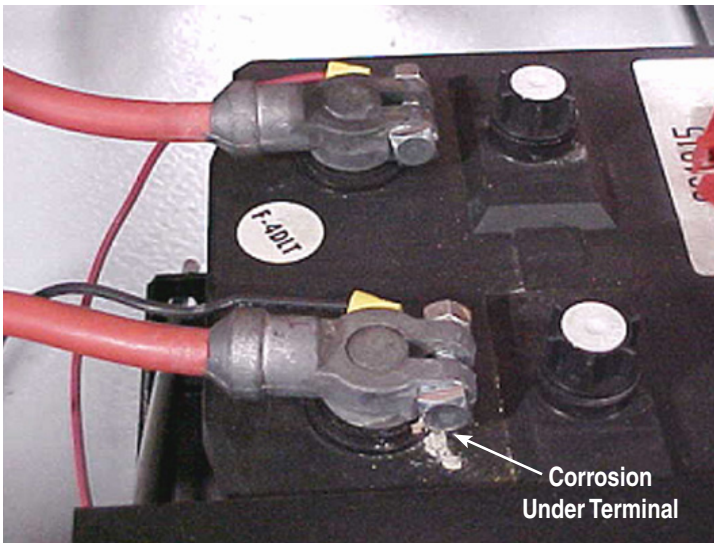
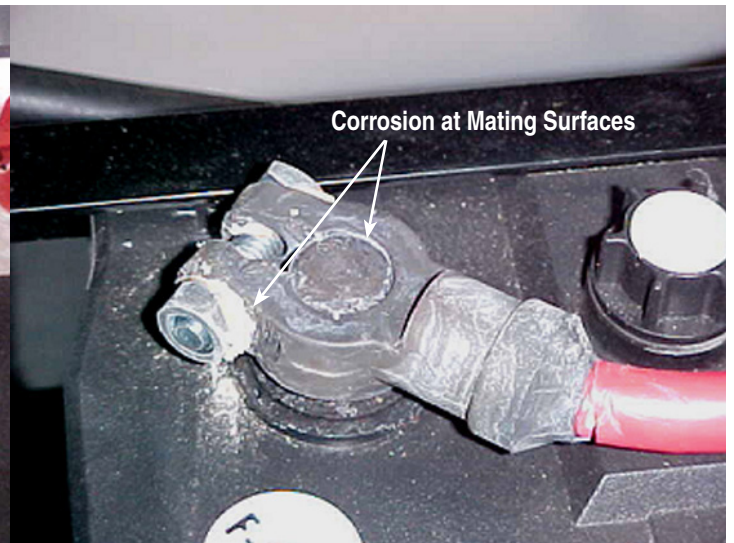
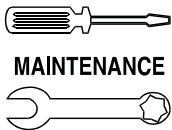


Figure 5 — Corrosion at Mating Surfaces





MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS Generac Power Systems

- Clean the battery top to eliminate conductive paths created by dried or wet electrolyte and to prevent corrosion.
- To prevent corrosion on terminals, thinly coat the terminals, terminal clamps and exposed metal around the battery with dielectric grease, high temperature wheel bearing grease or silicone. Do not use the felt or metal washers between the mating conductive surfaces with side, stud or "L" terminal batteries. Use of some stainless steel alloys and other metal washers, nuts and bolts have also been known to cause problems with electrolysis and high resistance.

Figure 6 — Ground Connections

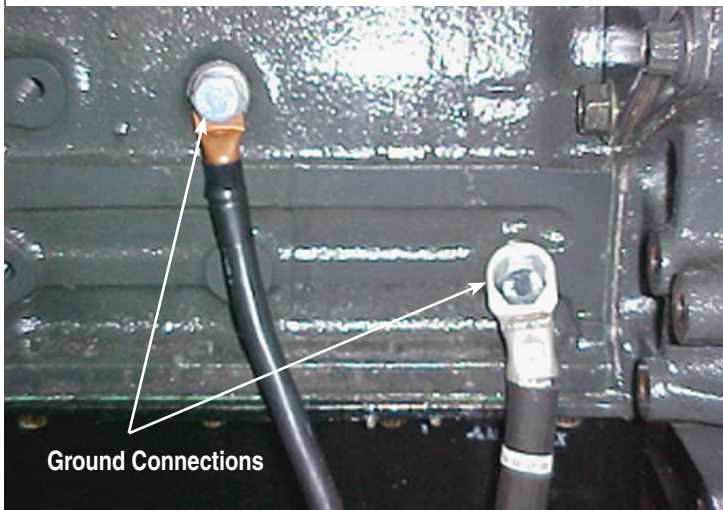


Figure 7 — Greased Connection



- Replace any battery cables (or cable terminals) that are corroding, swelling or damaged with equal or larger diameter cable and never trim strands to fit the cable into the cable terminal when replacing the terminal.

Corrosion is caused by one or more of the following:

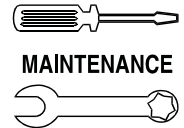
- Dirty or wet battery tops normally caused from expansion of electrolyte from overfilled cells
- Acid fumes leaking through the vent caps, which could be a sign of overcharging.
- Electrolysis due to the mismatch of metal alloys used in the battery posts and terminals.
- Clean the alternator or charging system to allow better heat transfer and check the alternator belts for cracks and correct tension.
- Replace the battery if the battery case is cracked or leaking.
- Battery Testing can be done in more than one way. The most popular is measurement of specific gravity and battery voltage. To measure specific gravity, use a temperature compensating hydrometer. To measure voltage, use a digital D.C. Voltmeter.
- You must first have the battery fully charged. The surface charge must be removed before testing. If the battery has been sitting at least several hours you may begin testing. To remove surface charge the battery must experience a load of 20 amps for 3 plus minutes.

<u>State of Charge</u>	<u>Specific Gravity</u>	<u>Voltage</u>
100%	1.265	12.7
*75%	1.225	12.4
50%	1.190	12.2
25%	1.155	12.0
Discharged	1.120	11.9

* Sulfation of Batteries starts when specific gravity falls below 1.225 or voltage measures less than 12.4 volts. Sulfation hardens the battery plates reducing and eventually destroying the ability of the battery to generate volts and amps.

The results of your testing should be as follows:

Hydrometer readings should not vary more than .05 difference between cells.



Digital Voltmeters should read as the voltage is shown in this document. If you have voltage readings in the 10.5 volts range on a charged battery, that indicates a shorted cell.

If you have a ***maintenance free wet cell***, the only ways to test are voltmeter and load test. Most of the maintenance free batteries have a built in hydrometer that tells you the condition of 1 cell of 6. You may get a good reading from 1 cell but have a problem with other cells in the battery.

Load testing is yet another way of testing a battery. Load testing removes amps from a battery much like starting an engine would. Some battery companies label their battery with the amp load for testing. This number is usually 1/2 of the CCA rating. For instance, a 500CCA battery would load test at 250 amps for 15 seconds. A load test can only be performed if the battery is near or at full charge.

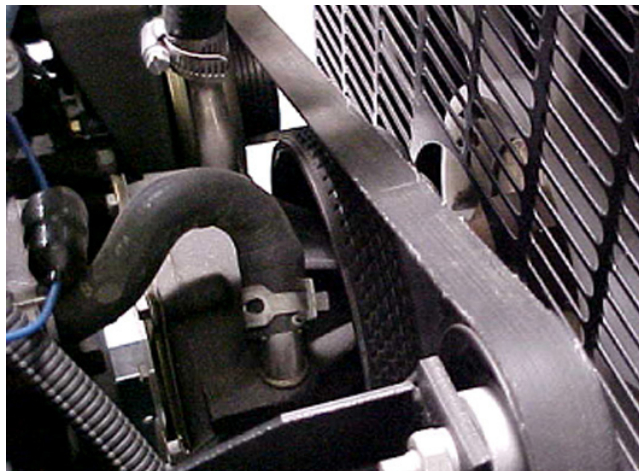
BELTS

People often wait until they get some indication that their engine needs service - such as a noise or squeal - before calling a service dealer. Although you should have a professional technician look at your belts and hoses as part of your regular maintenance schedule, basic inspection should be done by the owner. By conducting monthly inspections of the belts, the owner can help prevent premature engine wear and extend the life of the engine.

SERPENTINE BELTS

Definition: A type of flat rubber drive belt that is used to turn multiple accessories on the front of an engine. It is called a serpentine belt because of the way it snakes around the various pulleys. Many engines now have a single serpentine drive belt because it eliminates the need for several separate V-belts. A spring-loaded pulley maintains tension on the serpentine belt. This does away with the need to re-tension the belt when it is replaced. Serpentine belts generally last 25% to 50% longer than conventional V-belts.

Figure 8 — Serpentine Belt

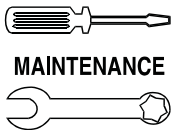


V-BELTS

V-belts are named for their appearance. They have a relatively narrow outside surface, then get thicker before tapering down. With V-belts, it is common for engines to have multiple belts driving the accessories.

BELT INSPECTION

- Check your engine's belts before you start the engine, while the engine is still cold. This will help avoid the possibility of an accidental burn from a hot engine component or an injury caused by a moving part of the engine.
- Carefully inspect the belts along their edges and undersides for any signs of wear (see "Visual Signs of Wear" below). These danger signs indicate a belt may need to be replaced or that a belt-driven component may be failing.
- Belt tension should be checked and adjusted on a regular basis. If the belt tension is too tight, it can cause bearings in the accessory components, and even in the engine itself, to wear prematurely. If it is too loose, the belt will slip and squeal, causing the accessory components to work less efficiently. Inadequate tension also will cause the belt to wear excessively. The most accurate way to check tension is with a belt tension gauge, but you can estimate correct tension by pressing on the belt along its longest straight section. If the tension is correct, the belt will only have about 1/2 inch to 1 inch of play.



MAINTENANCE

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- Drive belts are necessary to operate the accessory components attached to the engine, such as the alternator, and the water pump. To fully assess the condition of the belts, do a visual inspection and test the belt tension.

VISUAL SIGNS OF WEAR

- **Glazing:** The side or contact area of the belt becomes slick and shiny when a loose belt slips in the pulley. The glazed belt can no longer grip adequately and the belt slips even more.
- **Cause:** When in motion, the belt makes contact with an object in its path such as a flange or bolt. This may be caused by improper belt tension or pulley bearing size. Grease and oil on the pulley can also cause glazing.

Figure 9 — Glazed Belt



- **Improper Install:** A belt rib begins separating from the joined strands. If left unattended, the cover will often separate, causing the belt to unravel.
- **Cause:** Improper belt installation is a common cause of premature failure. One of the outer-most belt ribs is placed outside the pulley groove, causing a belt rib to run without a supporting or aligning pulley groove.

Figure 10 — Improper Install



- **Pilling:** The belt's rubber compound wears off and builds up on the drive pulleys (the wheel that is driven by or drives the belt).
- **Cause:** There are a number of causes, including lack of tension, misalignment, worn pulleys or a combination of these factors. Pilling is found most frequently in diesel engines, but is not isolated to them.

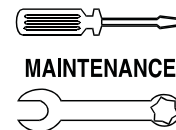


Figure 11 — Pilled Belt



- **Cracking:** Cracks occur because the belt is exposed to heat and stress.
- **Cause:** With continuous exposure to high temperatures, the stress of bending around the pulley leads to cracking. Cracks begin on the ribs and grow into the cord line. As a rule, if three or more cracks appear in a three-inch section of a belt, eighty percent of the life is gone and the belt should be replaced.

Figure 12 — Cracked Belt



- **Chunking:** Parts of the belt break off when cracks worsen.
- **Cause:** Chunking can happen when several cracks in one area move parallel to the cord line. Heat, age and stress are the primary contributors.

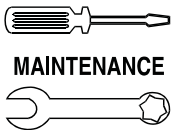


Figure 13 — Chunking



- **Uneven Rib Wear:** Belt shows damage to the side with the possibility of breaks in the tensile cord or jagged edged ribs. A thumping or grinding noise may also be heard when running.
- **Cause:** A foreign object in the pulley can cause uneven wear and cut into the belt.

Figure 14 — Uneven Rib Wear



- **Misalignment:** Sidewalls of the belt may appear glazed or the edge-cord may become frayed and ribs removed. A noticeable noise may result. In severe cases, the belt can jump off the pulley.
- **Cause:** Pulleys out of alignment due to either non-parallel shafts, or incorrect location on shafts. Misalignment forces the belt to kink or twist while running, causing premature wear.

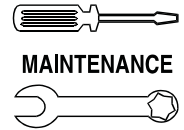


Figure 15 — Misalignment



BELT WEAR NOISE INDICATORS

- **Squealing:** Squealing is a continuous sound that often occurs when starting the engine after the engine has sat for awhile. It also can occur when a higher strain is put on the engine (such as after increasing the load). Squealing usually indicates belt wear. However, if enough water is splashed onto the drive belts, a squealing noise may be a normal condition associated with slippage due to the belt being wet.
- **Slapping:** A slapping sound can be caused by a loose belt or a belt misalignment.

FILTERS

AIR FILTERS

Air is necessary for successful combustion in your engine. In fact, for efficient combustion, a modern engine requires several thousand times as much air as it does fuel. Clean air - air almost 100% pure - is critical to engine survival and vital to its performance.

There are operational signs that an air filter has become completely plugged. The engine begins to lose power, and fuel consumption increases. Black smoke may blow from the exhaust stack. Continued operation with a plugged air filter may very well damage the engine.

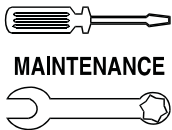
It is impossible to determine, just by looking, when air filters should be changed. An element that looks relatively clean may be almost totally plugged with ultra-fine particles from exhaust smoke or air pollutants.

On the other hand, a filter that looks dirty may still have many hours of useful life. Remember that until maximum acceptable restriction is reached, the accumulation of dirt in the filter actually adds to its efficiency.

Before disposing of old air filters, always inspect them carefully. Their appearance will tell you much about the performance of the entire air-induction system.

- An accumulation of black, oily soot might mean that the air intake is located too close to the exhaust. Consider relocation.
- An accumulation of dirt on the clean side of the filter element might indicate a split in the filter media. Also, determine if the end seal is being bypassed or if a gasket is leaking. Do not attempt to reuse the filter.
- Rust on the filter's metal parts can mean that water is being drawn in with the air. Again, check the location of the intake.

An Important Caution: Generac does not recommend the cleaning of air filter elements. Since all contaminants cannot be removed, service intervals become progressively shorter. Further, the cleaning process might damage the filter, leading to engine damage as well.



MAINTENANCE

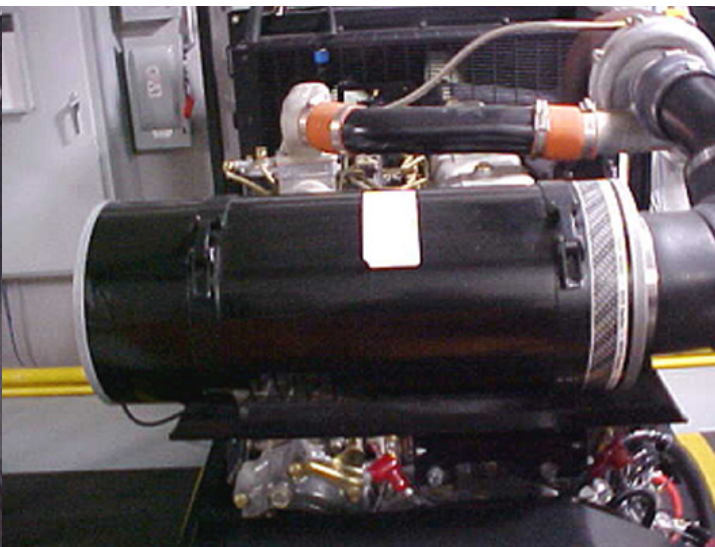
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Figure 16 — Automotive Type Air Filter



Figure 17 — Canister Type Air Filter



FUEL FILTERS

You've heard all the stories about contaminated fuels. No. 2 diesel fuel is one of today's most variable fuel sources. Its quality and purity range unpredictably from very good to very poor. Current API specifications allow "acceptable" levels of impurities (sulfur and wax) and contaminants (dirt, ash, and water). It's likely that these regulations will weaken even further in the future.

The deterioration of fuel is almost inevitable. Contaminants are introduced into the fuel system through mixing, transferring, and storage.

Micro-organisms can quickly become major problems.

Colonies of fungus and bacteria, both aerobic and anaerobic, feed on your fuel. Commonly known as humbugs, they spread rapidly in the presence of moisture. They ride along with the fuel, easily pumped from one tank to another. As the micro-organisms accumulate, they will spread through the fuel system and quickly plug the fuel filter. A coating of slime will develop over the entire surface of the media. Chemical treatment with a biocide is the only effective solution to the problem of bacterial and fungal growth.

Contaminants arrive in your fuel in two forms - precipitates and particulates. Precipitates are non-combustible materials formed when fuel oxidizes. They may also form if, as sometimes happens, two incompatible fuels are blended. Significant swings in temperature accelerate the precipitation problem. Because precipitates are generally heavier than fuel, they tend to settle to the bottom of tanks. Decanting or carefully drawing off the fuel will leave the precipitates behind. Particulates, sometimes called "asphaltenes," are black, tar-like contaminants. A residue of the refining process, particulates plug filters quickly. There is no known chemical treatment for the removal of these contaminants.

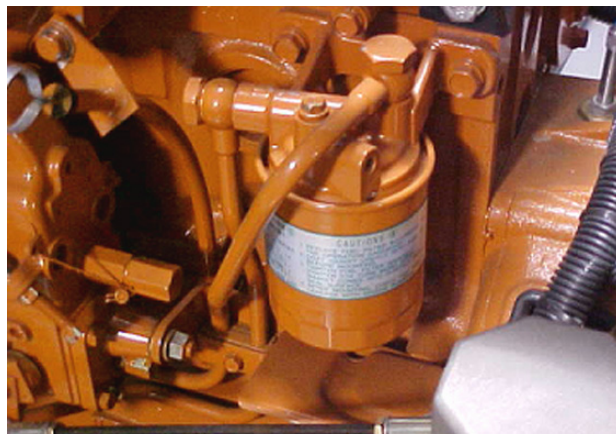
Wax, in moderation, is a desirable fuel component. It adds energy to the fuel. In operation during cold weather, however, control of wax becomes necessary, usually by applying additives or changing to a higher grade of diesel fuel. Like water, wax begins to thicken at colder temperatures. As wax gels, it can restrict and eventually stop fuel flow. Wax thickening involves a couple of critical temperature points:

- cloud point, the temperature at which solid crystals of wax begin to form - the fuel looks opaque, but it still flows.
- the pour point, the temperature at which the gel forms - the fuel cannot be poured.

Figure 18 — Fuel Filter



Figure 19 — Fuel Filter



After replacing the fuel filter, you will need to re-prime and bleed off the air from the fuel system using the following guidelines.

- Locate the fuel priming pump, this will be located between the incoming fuel line and the fuel filter. Some examples are shown in the following pictures.

Figure 20 — Fuel Primer

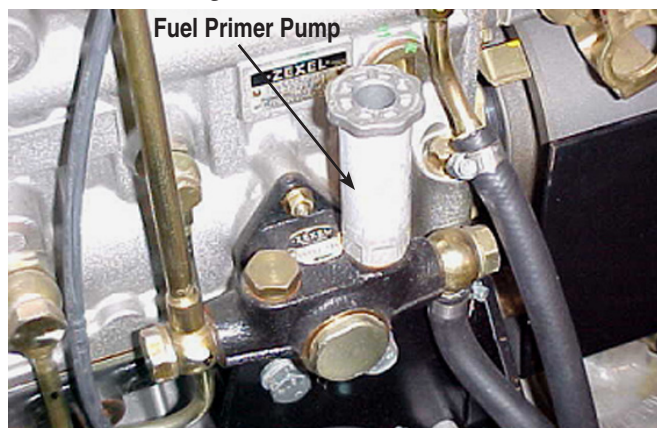


Figure 21 — Fuel Primer

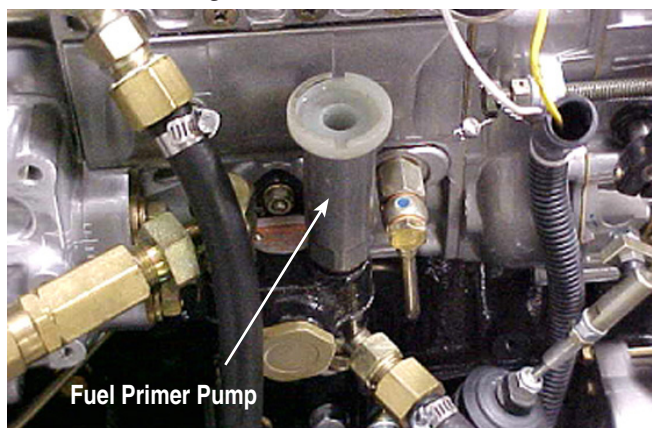


Figure 22 — Fuel Primer

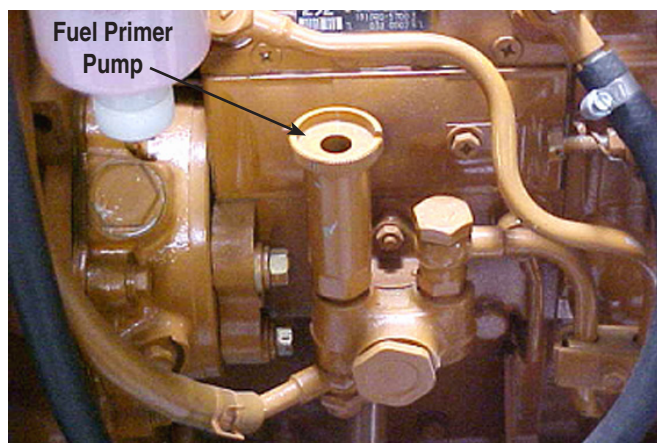
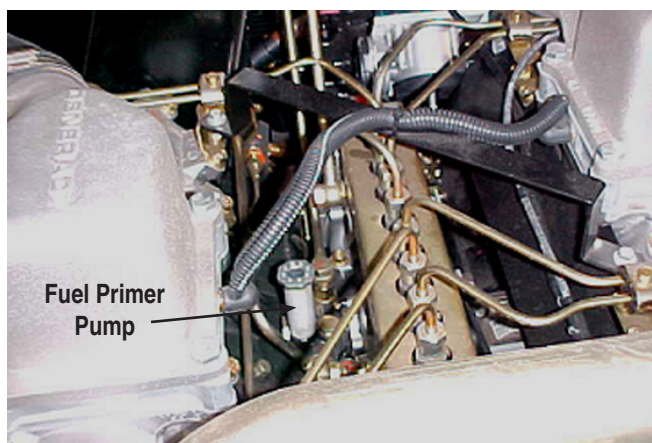
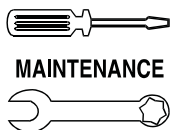


Figure 23 — Fuel Primer





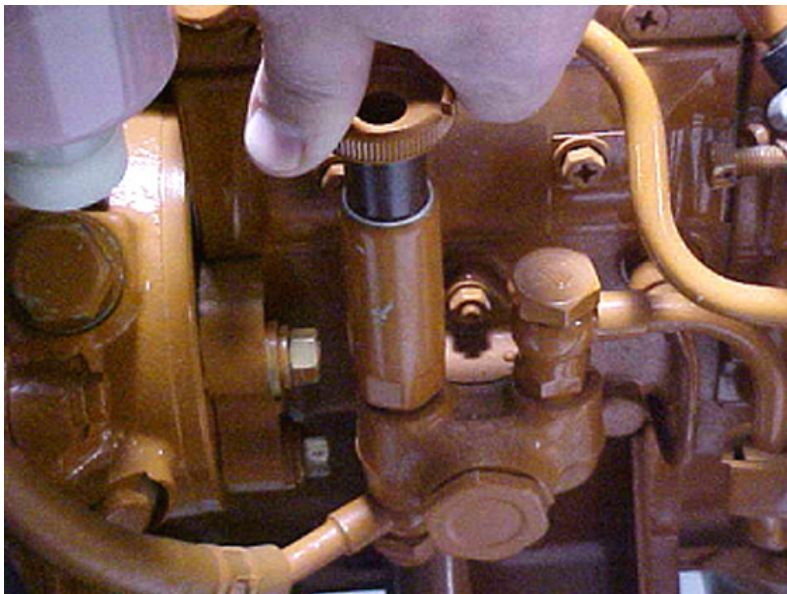
MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS

Generac Power Systems

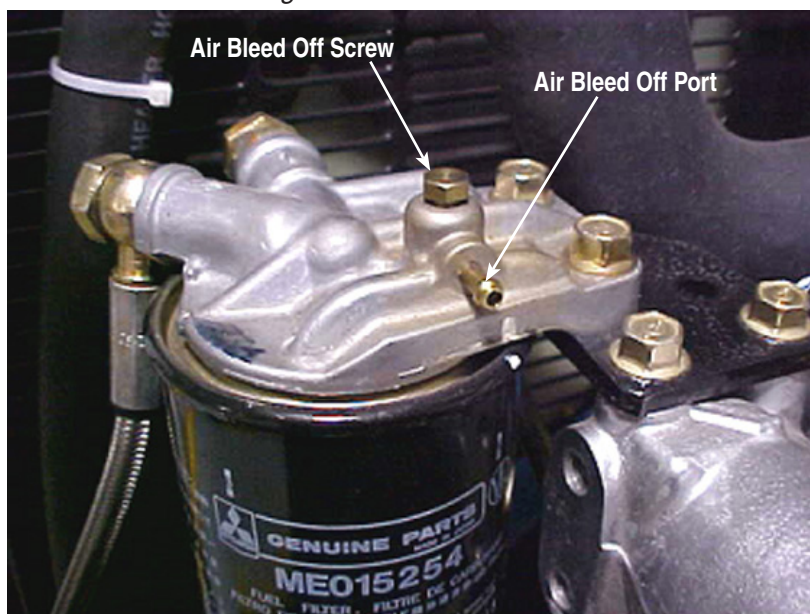
- Rotate the knob counter-clockwise to loosen, and pull the handle up.

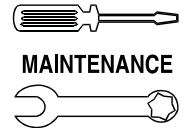
Figure 24 — Fuel Primer Pump



- Locate the air bleed off screw and the bleed off port.
- You will need to provide a container to catch any fuel which will be discharged during the following steps. A piece of clear tubing attached to the port and running down to a container on the ground will make this procedure easier.
- Loosen the air bleed off screw between 1 and 1 ½ turns.
- Start priming the system by moving the primer knob in an up and down motion.
- After a steady stream of fuel is being discharged from the port, tighten the bleed off screw.
- Push the primer knob down fully and rotate clockwise to tighten.
- Remove the hose from the port and clean any spills, dispose of any waste according to federal, state and local codes.

Figure 25 — Air Bleed Off



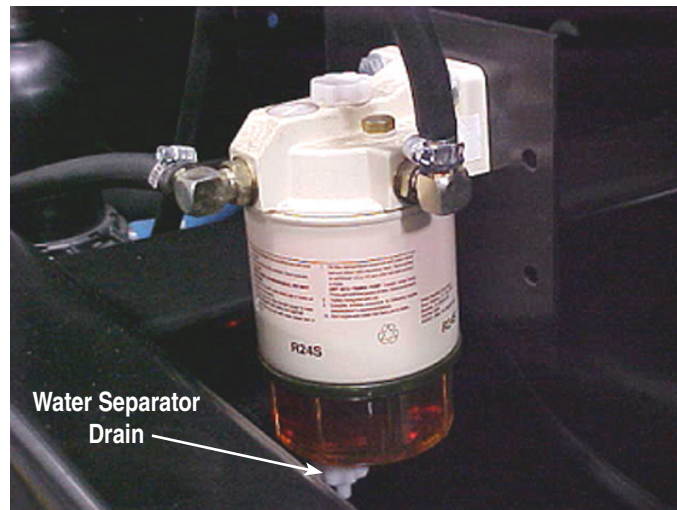


WATER SEPARATORS

Potentially the most damaging of the contaminants, water destroys the lubricative properties of your fuel. With lost lubricity, the fuel can scour pumps and even blow out injector tips. At colder temperatures, water in the fuel can freeze and contribute to total fuel stoppage. The good news is, water can be removed from fuel - within limits.

- To drain the water from the separator, place a container under the drain to catch the discharge, open the drain valve until all of the water is removed from the separator and then close the drain valve.
- Clean any spills and dispose of the waste in accordance with all federal, state and local codes.

Figure 26 — Water Separator



OIL FILTERS

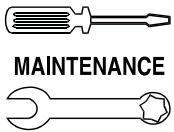
Modern oils play vital functions in protecting engines, especially in a heavy-duty situation. Lubricating oil acts to reduce friction and wear, cool engine parts, seal combustion chambers, clean engine components and inhibit corrosion. These functions are carried out by special additives in the oil, which complement the action of the oil itself. The protective action of the lubricating oil and its additives are supported and balanced by the work of the lube filter.

Lube filters, particularly those designed for heavy-duty applications, have the sole purpose of keeping damaging contaminants away from sensitive engine parts. Filters trap oil contaminants in two ways: Some particles adhere to filter media as the oil flows through the filter. Such particles attach themselves to the media surface without plugging up the media pores.

Other particles are trapped in the filter media by the pressure of the oil as it flows through the filter. As the oil changes direction in its path through the filter, particles are driven or impinged into the media. Ideally, most of these particles are trapped in the outer portion of the media, leaving inner media surfaces open to continue catching particles that slip through. Eventually, however, media pores will fill up and the filter begins to lose its effectiveness.

When changing the engine oil, always replace the engine oil filter(s).

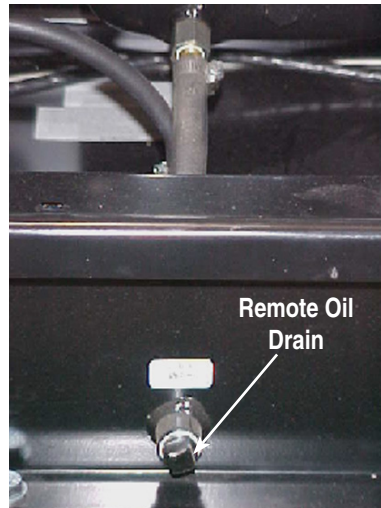
- Place a container in such a location as to catch the used oil and avoid spills.
- Remove the engine oil drain plug, or the remote oil drain plug and allow all of the used engine oil to drain out of the sump.
- Clean the drain plug, if the plug had teflon sealer on the threads, clean off the old teflon and place new teflon sealer on the clean threads.
- Replace the drain plug, do not over tighten.
- Move the container that has the used engine oil in it under the location of the oil filter(s) to catch any spills that occur when the oil filters are removed, or place some oil absorbent mats or rags under the area of the oil filter(s).
- Remove the used engine oil filter(s), being careful not to spill any oil remaining in the filter(s) and place the filter(s) in a container.



MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS Generac Power Systems

Figure 27 — Remote Oil Drain



- Clean all mating surfaces and lightly coat the gasket on the oil filter(s).
- Tighten the oil filter until the gasket contacts the mating surface, then tighten another $\frac{3}{4}$ turn.

Figure 28 — Oil Filters

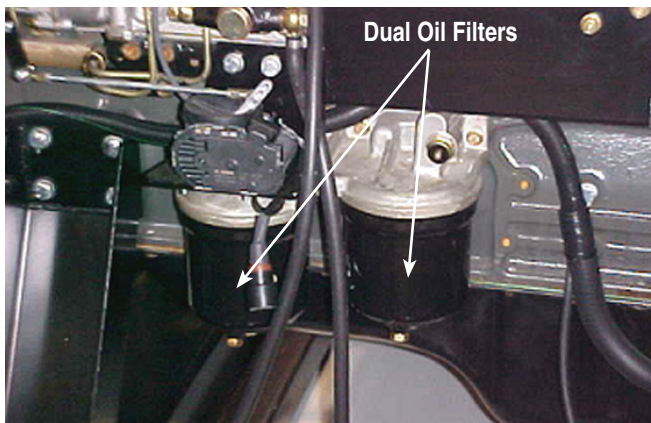


Figure 29 — Oil Filter

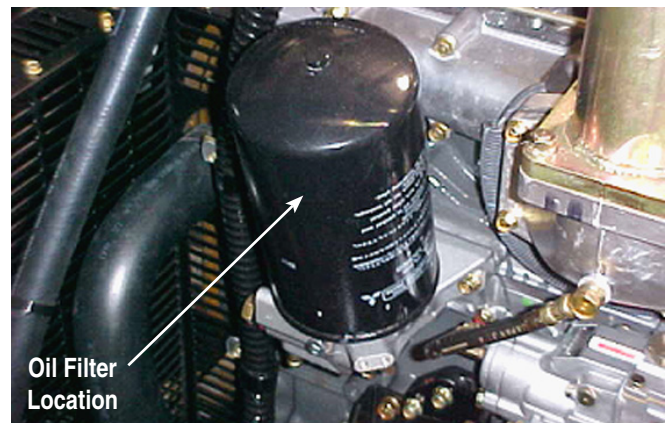


Figure 30 — Oil Filter

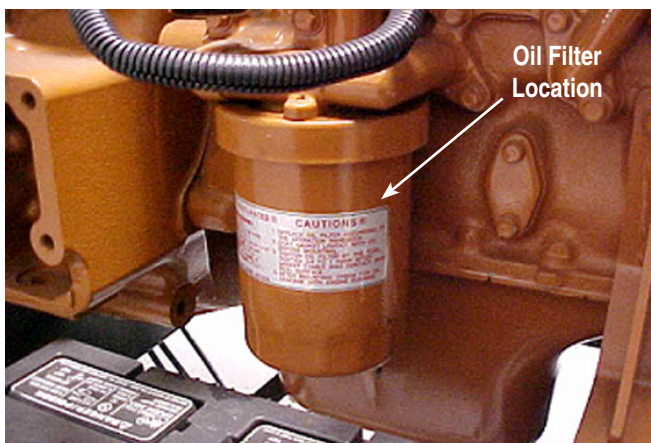
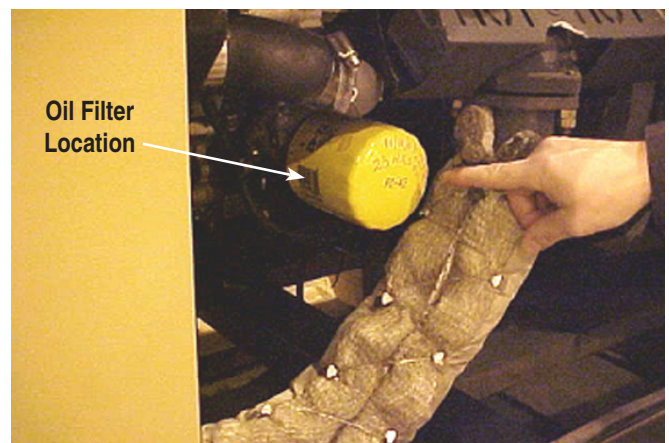
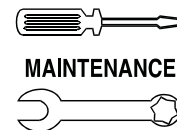


Figure 31 — Oil Filter





- Locate the oil fill cap, this is usually located on the valve cover, and remove the cap.
- Add the new engine oil to this location (refer to the owner/operator manual or the engine service manual for the correct grade/viscosity rating and oil capacity of the engine).

Figure 32 — Oil Fill Cap

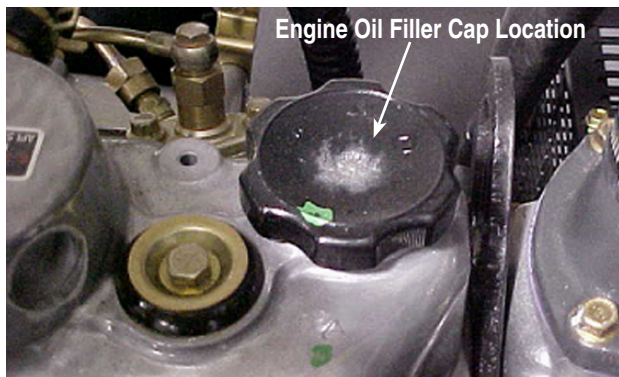


Figure 33 — Oil Fill Cap



Figure 34 — Oil Fill Cap



- After adding the correct amount of oil, check the level indication on the dipstick, this may read high at this time due to no oil being in the new filter you just installed.
- Replace the oil filler cap.
- Start the engine and verify that there are no leaks.
- Shut down the engine, and recheck the level indication on the dipstick, adjust as necessary.
- Clean up the area of any spills and dispose of all materials according to federal, state and local codes.

Figure 35 — Oil Fill and Dipstick

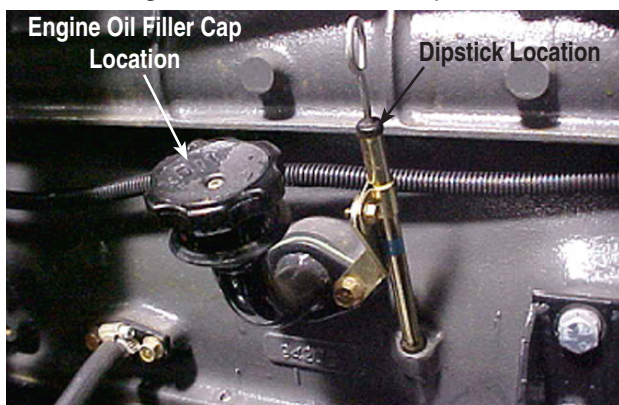
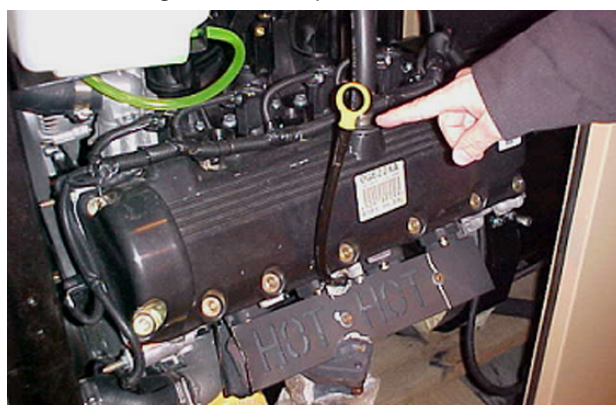
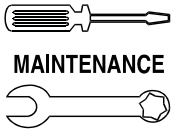


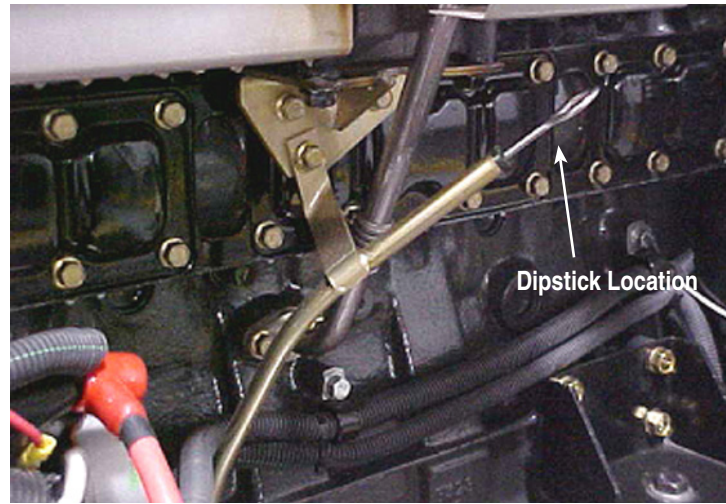
Figure 36 — Dipstick Location





MAINTENANCE

Figure 37 — Dipstick



SPARK PLUGS

The spark plug is quite simple in theory: It forces electricity to arc across a gap, just like a bolt of lightning. The electricity must be at a very high voltage in order to travel across the gap and create a good spark. Voltage at the spark plug can be anywhere from 5,000 to 100,000 volts, depending on the type of ignition system being used.

The spark plug must have an insulated passageway for this high voltage to travel down to the electrode, where it can jump the gap and, from there, be conducted into the engine block and grounded. The plug also has to withstand the extreme heat and pressure inside the cylinder, and must be designed so that deposits do not build up on the plug.

Spark plugs use a ceramic insert to isolate the high voltage at the electrode, ensuring that the spark happens at the tip of the electrode and not anywhere else on the plug; this insert does double-duty by helping to burn off deposits. Ceramic is a fairly poor heat conductor, so the material gets quite hot during operation. This heat helps to burn off deposits from the electrode.

Some engines require a hot plug. This type of plug is designed with a ceramic insert that has a smaller contact area with the metal part of the plug. This reduces the heat transfer from the ceramic, making it run hotter and thus burn away more deposits. Cold plugs are designed with more contact area, so they run cooler.

RESISTOR PLUGS

When the spark jumps the spark plug gap it causes a high frequency burst of energy, that is known as radio frequency interference (RFI). Placing a resistor within the spark plug suppresses the RFI. Without the resistor plugs in your engine, the RFI could cause interference with the engine/generator's electronics.

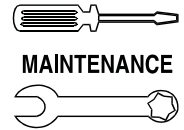
HEAT RANGE

The most basic function of the spark plug is to ignite the air/fuel mixture. Voltage must be supplied by the ignition system to cause a spark to jump across the spark plug's gap and ignite the air/fuel mixture.

To survive the combustion chambers high temperatures and function properly, the spark plug must dissipate the heat that it absorbs. The temperature of the spark plug's firing end must be kept low enough to prevent pre-ignition, but high enough to prevent fouling. It is important to remember that spark plugs do not create heat, but instead they must remove heat. The heat range determines the plug's ability to dissipate the heat from the spark plug to the cylinder head where it is absorbed by the coolant system.

How quickly this heat is transferred is determined by:

- The insulator nose length.
- Insulator nose surface area exposed to the air/fuel mixture.
- The construction of the electrode and the porcelain insulator.



Heat flow path of a Spark Plug

The insulator nose length is the distance from the tip of the insulator to the point where insulator meets the metal shell. Since the insulator tip is the hottest part of the spark plug, the tip temperature is a primary factor in pre-ignition and fouling. The spark plug tip temperature must remain between 500°C-850°C. If the tip temperature is lower than 500°C, the insulator area around the center electrode will not be hot enough to burn off combustion chamber deposits. If the tip temperature is higher than 850°C the spark plug will overheat which may cause the ceramic around the center electrode to blister and the electrodes to melt. In identical spark plug types, the difference from one heat range to the next is the ability to remove approximately 70°C to 100°C from the combustion chamber.

Tip Temperature and Firing End Appearance

There are three basic spark plug operating conditions:

- Fouled operating under 500 C
- Overheated operating over 800 C
- Good operating between 500 C and 800 C

The insulator nose length is a large factor in determining the heat range of a spark plug, the longer the insulator nose, the less heat is absorbed, and the further the heat must travel into the cylinder head water journals. This means the plug has a higher internal temperature, and is said to be a hot plug. The heat range of a plug has no relationship to spark quality or intensity. Many people mistakenly associate a hotter plug with a spark plug that has a hotter spark. A cold spark plug has a shorter insulator nose and absorbs more combustion chamber heat. This heat travels a shorter distance, and allows the plug to operate at a lower internal temperature. The colder type removes heat more quickly, and will reduce the chance of pre-ignition/detonation and melting or damage to the firing end.

What affects the spark plugs operating temperatures?

Below is a list of some of external influences on a spark plug's operating temperatures. The following symptoms or conditions may have an effect on the actual temperature of the spark plug. They do not affect the heat range but how the plug is performing under the current conditions. Generac considers these parameters when determining the characteristics of the spark plug required. The spark plug cannot create these conditions but it must be able to cope with them. Air/Fuel mixtures seriously affect engine performance and spark plug operating temperatures.

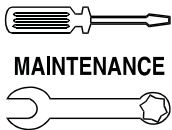
- Rich air/fuel mixtures cause tip temperature to drop, causing fouling and poor performance.
- Lean air/fuel mixtures cause plug tip and cylinder temperature to increase, resulting in pre-ignition, detonation, and possibly serious spark plug and engine damage.
- Forced Induction will elevate spark plug tip and in-cylinder temperatures.
- Engine Load.
- Increases in firing-end temperature are proportional to engine speed and load.

Humidity

- As humidity increases, air intake volume decreases
- Result is lower combustion pressures and temperatures, causing a decrease in the spark plug's temperature and a reduction in available power.
- Air/fuel mixture should be leaner, depending upon ambient temperature.

Barometric Pressure/Altitude

- Also affects the spark plug's tip temperature
- The higher the altitude, the lower cylinder pressure becomes. As the cylinder temperature decreases, so does the plug tip temperature
- Many mechanics attempt to "chase" tuning by changing spark plug heat ranges
- The real answer is to adjust jetting or air/fuel mixtures in an effort to put more air back into the engine.



TYPES OF ABNORMAL COMBUSTION

Pre-ignition

- Defined as: ignition of the air/fuel mixture before the pre-set ignition timing mark.
- Caused by hot spots in the combustion chamber...can be caused (or amplified) by too hot a spark plug, lean air/fuel mixture, or insufficient engine cooling.
- A change to a colder plug, or a richer fuel mixture may be in order.
- You may also need to check the engine's cooling system.
- Pre-ignition usually leads to detonation; pre-ignition and detonation are two separate events.

Detonation

- The spark plug's worst enemy.
- Can break insulators or break off ground electrodes.
- Pre-ignition most often leads to detonation.
- Plug tip temperatures can spike to over 3000°F during the combustion process.
- Most frequently caused by hot spots in the combustion chamber.

Hot spots will allow the air/fuel mixture to pre-ignite. As the piston is being forced upward by mechanical action of the connecting rod, the pre-ignited explosion will try to force the piston downward. If the piston can't go up (because of the force of the premature explosion) and it can't go down (because of the upward motion of the connecting rod), the piston will rattle from side to side. The resulting shock wave causes an audible pinging sound. This is detonation.

- Most of the damage that an engine sustains when "detonating" is from excessive heat.
- The spark plug is damaged by both the elevated temperatures and the accompanying shock wave, or concussion.

Misfires

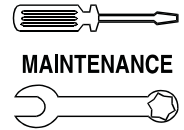
- A spark plug is said to have misfired when enough voltage has not been delivered to light off all of the fuel present in the combustion chamber at the proper moment of the power stroke.
- A spark plug can deliver a weak spark for a variety of reasons...defective coil, too much compression with incorrect plug gap, dry fouled or wet fouled spark plugs, insufficient ignition timing, etc.
- Slight misfires can cause a loss of performance for obvious reasons (if fuel is not lit, no energy is being created).
- Severe misfires will cause excessive fuel consumption, poor performance, and can lead to engine damage.

Fouling

- Will occur when spark plug tip temperature is insufficient to burn off carbon, fuel, oil or other deposits.
- Will cause spark to leach to metal shell...no spark across plug gap will cause a misfire.
- Wet-fouled spark plugs must be changed.
- Dry-fouled spark plugs can sometimes be cleaned by bringing engine up to operating temperature.

Changing Spark Plugs

- Always change your spark plugs with the engine cold. Grabbing the plug wire by the boot, carefully pull the spark plug wire from the end of the spark plug. Do not pull the wire itself. If the boot sticks, twist the boot left and right and pull the plug wire off. I would recommend changing the plugs one at a time to avoid mixing up the spark plug wires.
- If you have it, use compressed air to blow any dirt away from the spark plug area. Otherwise, clean off the old plug and the area around it with a rag or small brush. This will help prevent any foreign material from falling into the cylinder when the plug is removed.
- Remove the plug by turning it counterclockwise with a spark plug socket and ratchet. Once you crack it loose, spin it out about three or four turns. Then remove the socket and remove it completely by hand. If you can't reach it, slip a piece of 5/16" vacuum line over the spark plug and turn it out with that.
- If you haven't done so yet, gap the new plug with a spark plug gap gauge (The proper gap can be found in the owner/operators manual that came with the unit). Slip the correct thickness wire or feeler between the inner and outer electrodes at the tip of the plug. A flat gauge is good; the wire type is better. When the plugs are properly gapped, the wire or feeler should slide between the electrodes with a slight drag. If the gap is incorrect, gently bend the outer electrode slightly until the correct gap is achieved. Make sure that the outer electrode is centered directly over the inner electrode. If it's not, align the two by gently bending the outer electrode.



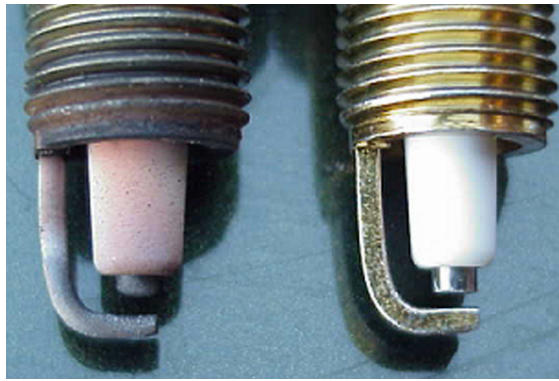
- Take a good look at the cylinder head threads. They should be in good condition, clean, and free of dirt and debris. This new spark plug should freely screw into the cylinder head by hand. Any binding of the plug is an indication that there's a problem. Remove the plug and inspect the threads.
- Insert the plug into the spark plug hole by hand and turn it clockwise until it's snug. I always attach a short piece of 5/16" rubber hose to the top of the plug and use it as an extension to install the new plugs.
- After installing the plug by hand as far as it will go, firmly tighten it with a spark plug wrench or socket. It's a good idea to use a torque wrench, if one is available, to ensure that the plug's properly seated. Be very careful; do not over tighten the spark plugs. Remember, you'll get an accurate torque reading only if the spark plug and cylinder head threads are clean and dry.
- Reattach the plug wire to the new plug. Use a twisting motion on the boot until it's firmly seated on the top of the plug. You will feel and hear a click as the wire clamps onto the spark plug.

Repeat these same steps for the other plugs. If you take your time this is a fairly easy job and will probably take you about an hour, depending on how many plugs your engine has.

Diagnosing spark plugs is really pretty straight forward, based on the appearance of the electrode end of the spark plug. The following illustrations and descriptions should aid in helping you diagnose if there is an issue that is effecting the combustion process of the engine.

- Normal plug condition, note the difference in the gaps, the plug on the left has reached the end of it's useful life, also note the grounding electrode coloration, this is how a used spark plug should appear when it is removed. If any plugs do not appear like the plug on the left, contact your service dealer.

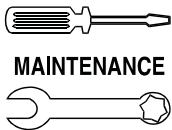
Figure 38 — Normal Plug Conditions



- Mechanical damage is caused by foreign objects in the combustion chamber or an improper plug reach where it contacts the piston. Even a piece of carbon can do this.
- To solve this, make sure you have the correct length tip spark plug as well as removing any foreign materials in the combustion chamber. In some cases you may have excessive carbon buildup on the backs of the intake valves that will have to be addressed.

Figure 39 — Mechanical Damage





MAINTENANCE

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- Detonation, in cases of severe detonation, insulators may become cracked or chipped. Improper spark plug gap settings will also cause the insulator tip to crack or chip.
- Detonation is tricky ... make sure that you verify correct ignition timing. Next check for an inoperative EGR system (if equipped) as well as proper function of the Knock Sensor (if equipped). Also, you will want to make sure you are using the correct heat range plug.

Figure 40 — Detonation



- Overheating, on this symptom you will notice a chalky appearance, white insulator, rapid electrode wear as well as an absence of deposits. The actual shell may also be discolored.
- To cure this you must first verify that the plug is the correct heat range, the ignition timing settings are correct, the air/fuel mixture is not too lean, there are no vacuum leaks and that the EGR valve (if equipped) is functioning properly.

Figure 41 — Overheating



- Oil fouled is an oily coating caused by poor oil control. Oil is leaking past worn valve guides, piston rings, or on some engines a possible intake gasket leak and then entering the combustion chamber.
- Check for worn valve guides, intake gasket sealing alignment, as well as worn cylinder walls and piston rings. A leak down test is a good place to start for what is causing this.

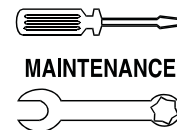


Figure 42 — Oil Fouled

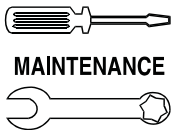


- Initial pre-ignition will usually look as a melted center electrode and/or ground electrode.
- Check for incorrect heat range plug, over-advanced timing, lean fuel mixtures, inoperative EGR valve or Knock Sensor (if equipped) and also look for hot spots or deposit accumulation inside the combustion chamber.

Figure 43 — Initial Pre-ignition



- Sustained pre-ignition, well this will be pretty obvious ... melted and/or missing center and/or ground electrodes as well as a destroyed insulator.
- Check for incorrect heat range plug, over-advanced timing, lean fuel mixtures, inoperative EGR valve or Knock Sensor (if equipped) and also look for hot spots or deposit accumulation inside the combustion chamber.
- After you see this, you'd better look for possible internal engine damage as well. (pistons, cylinder walls, valves, rings, etc.)



MAINTENANCE

BASIC MAINTENANCE INFORMATION FOR GENERAC STANDBY GENERATORS

Generac Power Systems

Figure 44 — Sustained Pre-ignition



- Carbon fouled is a very common visual condition on engines. Soft, black, sooty, dry-looking carbon. This indicates a rich mixture, weak ignition or wrong heat range plug (too cold).
- You will first need to verify plug heat range. Check choke as well as choke pull-off (if equipped) for proper function and adjustment. As a general rule on all computer-controlled engines, you need to also make sure that all input signals to the ignition module are working and accurate. This includes, but is not limited to, all temperature and pressure sensors as well as the system components. Lastly on all engines, check for vacuum leaks and weak spark or low voltage output.

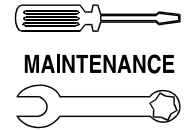
Figure 45 — Carbon Fouled



- Cracked or broken insulator, this is typically a result of improper installation or careless handling of the spark plug and should be replaced.

Figure 46 — Cracked Insulator





COIL-ON-PLUG IGNITION SYSTEMS

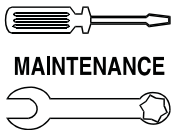
First it was the distributor that vanished. Now plug wires are starting to disappear. Spark plug wires are going away for the same reason that distributors went away. Engine manufacturers want to reduce costs and improve ignition performance and reliability. Plug wires are an assembly line nuisance, and are often the weak link in today's distributorless ignition systems. The plug wires must carry anywhere from 5,000 up to 40,000 or more volts to fire the plugs. This requires heavy insulation plus the ability to suppress electromagnetic interference (EMI). The wires must also be coated with a tough outer jacket to withstand high temperatures and chemical attack.

As reliable as today's plug wires are, there is always the potential for trouble. Even the toughest insulation can burn if a wire rubs up against a hot exhaust manifold. The connection inside the spark plug boot between the wire and plug terminal can also be damaged if someone jerks on the wire to remove the boot when changing sparkplugs. Plug wires can also radiate magnetic fields that may affect nearby sensor wires or other electronic circuits.

Attaching the ignition coils directly to the spark plugs eliminates the need for separate high voltage wires along with their potential for trouble. Eliminating the individual plug wires also eliminates the need for wire looms and heat shields. That's why coil-on-plug ignition systems are being used on a growing number of late model engines.

Detailed specifications for the engine you are servicing are essential, because there are many different ignition systems using coil packs, coil near plug or coil on plug. General Motors, for example, uses multiple coils with at least four different ignition control systems. Here are some general troubleshooting tips that can be helpful.

- Always verify the complaint. However, remember that no start, misfire and engine stalling complaints can be fuel related, spark plug related or both. These types of problems also cause high-hydrocarbon (HC) emissions.
- Check the basics. Make sure all wiring connections are good. Look for obvious problems like damaged wires, corrosion, etc. Make sure the charging system is normal. A weak charging system reduces the voltage to the primary side of the coil, which then prevents the induction of a powerful secondary current. Although the electronic ignition system will operate with as little as 7 volts input to the primary side, the voltage should be normal system voltage (12-14 volts) to create a strong spark.
- Coils can die, but they are more likely to be murdered. If you identify a defective coil, try to determine what caused the coil failure and correct the problem. Otherwise, the replacement unit will be doomed to failure just like the original coil. For example, a bad current limiting circuit in the ICM can cause rapid overheating and early ignition coil failure.
- If a coil is suspect, you may be able to swap it with another coil on the same engine to see if the misfire follows the suspected bad coil. If the misfire moves with the suspected coil, the unit is bad. If the misfire stays at the same cylinder, there may be a problem with the voltage supply or trigger signal to that cylinder. However, on some engines, swapping coils may be too time consuming.
- An ohm meter can be used to test coil winding resistance. Primary-side resistance, from coil minus to coil plus, is typically between 0.3 and 1.0 ohm on electronic ignition type coils.
- Secondary resistance values vary widely so consult a specifications chart for the engine you are servicing. If a spec chart isn't available, compare secondary readings among all the coils to see if any one is higher than the others. A high resistance indicates deterioration in the wiring. Surprisingly, a coil with high resistance may still fire the spark plug, but the voltage produced will be higher because the current must jump the open wiring in addition to jumping the spark plug gap.
- Although coils can generate up to 45 kV (45,000 volts), normal firing voltage is typically 20-25 kV. Voltage levels much over 25 kV indicate an opening in the coil windings, worn plugs or a bad connection between the coil and the plug. Prolonged exposure to more than 25 kV can cause a coil to fail by breaking down the internal insulation or overheating the unit.
- Shorted windings inside the coil can be difficult to diagnose. Just a few turns shorted together can cut maximum coil output from 45 kV to only 10 kV. Typically, a shorted coil will produce an adequate spark under light loads, but will misfire under load or hard acceleration.
- A coil may operate normally when the engine is started, but then fail after becoming heated. The engine will start and run well, but then develop an intermittent or steady misfire in one cylinder. The engine will operate normally during low-speed operation, but misfires under load or hard acceleration, especially after the engine has been running for a period of time. If possible, swap coils and see if the misfire follows the suspected coil.
- If an engine starts normally, but then develops random misfires, stalls or has other performance problems traced to the ignition system, and no problems are found in the secondary ignition system, check the crankshaft position sensor's and camshaft position sensor's operation. Make sure all connections between the sensors and the ICM are clean and tight.



MAINTENANCE

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Figure 47 — Coil on Plug

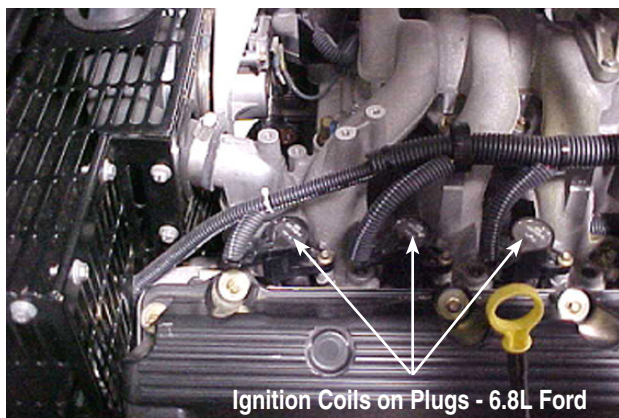
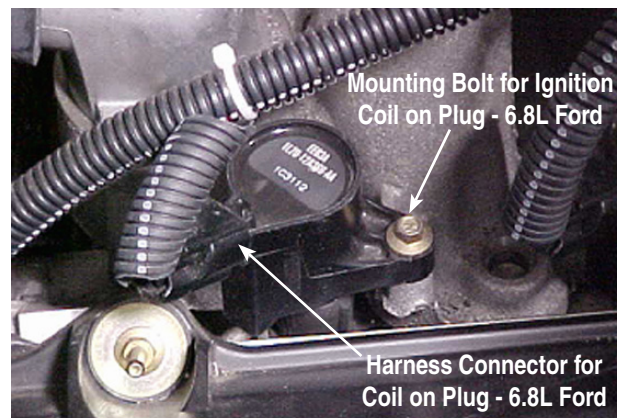
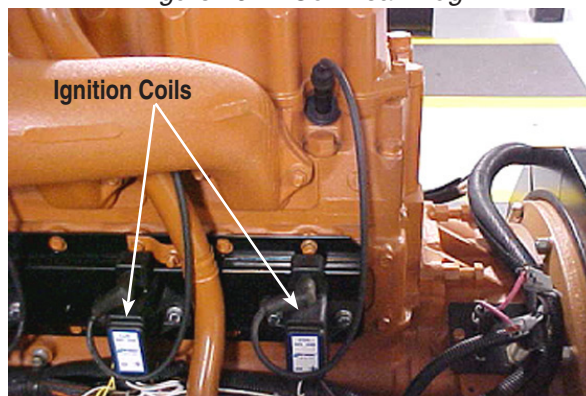


Figure 48 — Coil on Plug



- Coil near plug design, this is due to the proximity of the exhaust manifold to the plug location, eliminating the ability to locate the coil directly over the plug. Note the individual coils per plug and the short length of plug wire required.

Figure 49 — Coil Near Plug



HOSES

Hoses are used on many components of every engine. They carry liquids (diesel, LPL, and coolant,) and gases (LPG, and natural gas).

HOSE INSPECTION

Cold Engine Inspection

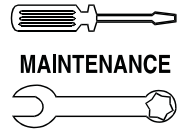
- Check for small leaks before you start the engine, while your engine is still cold. If your engine is operated when it is low on coolant, serious engine damage may occur. Replace the hose as soon as possible.
- Firmly squeeze the radiator hose. A hose that feels very hard or makes a "crunch" as it is squeezed is deteriorating and should be replaced. You also should replace hoses that are extremely soft, sticky or oil-soaked, as they are likely to fail.

Warm Engine Inspection



⚠ Caution should be used around hot engine parts to avoid accidental burns.

- After operating the engine long enough for the engine to be fully warmed up, turn off the engine. Make sure the upper radiator hose is not collapsed. If it is, it may indicate a problem and should be checked by a qualified service technician.



- Look carefully at all the hoses to spot any swollen areas. Such areas indicate weak spots.
- Inspect around hose ends for dampness, deposits or buildup of dried coolant. These could indicate that a "cold leak" is present. A cold leak is usually a very minor leak that occurs after the engine has cooled down, and expansion and contraction of components has occurred. All coolant leaks should be addressed as soon as you notice them. Over time, these leaks will get worse and could lead to an expensive repair. An inexpensive gasket or a simple tightening of a hose clamp may be all that is needed to avoid an expensive repair.

When To Replace Your Hoses

- When you replace a water pump
- When you replace a thermostat
- Every four years of service
- Anytime you notice damage or leakage

VISUAL SIGNS OF WEAR

- **Cracking:** Cracking is caused by heat and age, exposure to ozone, etc.
 - **Cause:** Increased ozone, caused by pollution, attacks bonds in certain rubber compounds. Tiny cracks occur primarily where the hose is stressed- at curves, bends and clamping locations. These cracks allow airborne contaminants to invade and weaken the hose.

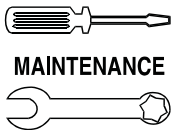
Figure 50 — Cracking



- **Oil Damaged:** Damaged hose is soft, gooey, or spongy to the touch. Bulges and swelling are readily apparent.
 - **Cause:** Oil reacts chemically with hose compounds to weaken the structure of molecular bonds. This causes the hose to soften, swell, and separate by layers, leading to certain failure. Oil can attack both external and internal surfaces of the hose.

Figure 51 — Oil Damage





- **Electrochemical Degradation:** ECD (Electrochemical Degradation) is not obvious by visual inspection. However, by squeezing a cool hose near the end between your thumb and fingers, you can actually feel voids, cracks or weak spots. Also, bulging may occur near clamped connections.
- **Cause:** ECD is a primary cause of most coolant hose failures. Modern engine materials react with the coolant to cause an electrical charge. This electrical current creates tiny internal cracks, which weakens the hose.

Figure 52 — Electrochemical Degradation



- **Heat Damage:** As with ECD, heat damage can occur internally and may not be easily detected by physical appearance. Swelling is one sure external sign of internal damage. External heat damage is often easier to detect since it can result in hardened and cracked hose covers.
- **Cause:** Overheating can cause reinforcement fibers within a hose to deteriorate. Engine heat, low coolant levels and/or temperature spikes all contribute to hose damage.

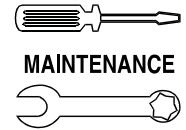
Figure 53 — Heat Damage



- **Leakage:** Moisture, seepage or drips form on or around clamps or connectors.
- **Cause:** Other than insufficient clamp torque, leakage is usually caused by the deteriorated condition of the hose and/or connector.

Figure 54 — Leakage





COOLANT

THE FUNCTIONS AND MAINTENANCE OF ENGINE COOLANT

The great importance of engine coolant to trouble-free engine operation is poorly understood. Maintenance managers and industry experts estimate that over 40% of a heavy-duty engine's downtime is related to coolant system problems, virtually all of which is preventable with proper preventative maintenance! The coolant removes excess heat from the engine. Without coolant, the engine metals would quickly soften and deform, resulting in catastrophic damage. In addition, since water is corrosive to engine metals, the coolant has to contain anti-corrosive protective chemicals to prevent rust and other damage that weakens the engine parts or radiator. There are many metals to protect: aluminum, steel, cast iron, copper, brass and solder. Furthermore, the coolant must not attack the rubber hoses or gaskets in the engine. Of course, it must provide antifreeze and anti-boil protection.

Antifreeze is actually a treatment for the water that is used to cool an engine. The water is very important, as it is the primary part of the coolant that removes the heat. The antifreeze treats the water to prevent rust, corrosion, hose failures, radiator failures, liner pitting and a host of other possible problems. The industry has agreed that most antifreeze/coolants are formulated to be mixed half-and-half with water. This is important, because even if you live in Minneapolis or Phoenix you should use 50% antifreeze to insure that your engine is properly protected. In extremely cold climates, it is permissible to increase the antifreeze to a maximum of 70%. Most people don't realize that above that, the freeze point actually gets warmer! The maximum antifreeze protection with EG (ethylene glycol) based antifreezes is -86 degrees F when 70% EG based antifreeze is mixed with 30% water.

GENERAL CONSIDERATIONS:

In most cases, local drinking water is just fine for use with EG based antifreeze. (We don't recommend well water or really hard water.) At every oil change, check the coolant to insure it is still at a 50/50 mixture with a test strip or refractometer.

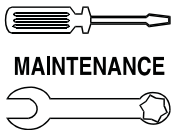
Use a mixture of half low silicate, ethylene glycol base antifreeze or propylene glycol base antifreeze and half soft water, or use a pre-mix antifreeze. Use only soft water or de-ionized water and only low silicate antifreeze. Do not use water that has been softened using a water softener filled with either salts or chlorides. If desired, add a high quality rust inhibitor to the recommended coolant mixture. Use only heavy-duty or all-duty formulation (fully formulated) coolants. Any high quality heavy-duty ethylene glycol antifreeze can be mixed with any other high quality heavy-duty ethylene glycol antifreeze, and any high quality heavy-duty propylene glycol antifreeze can be mixed with any other high quality heavy-duty propylene glycol antifreeze regardless of the color or brand of the antifreezes. When changing from ethylene glycol based antifreeze to propylene glycol based antifreeze or vice-versa, be sure that the engine and cooling system is thoroughly flushed of all remaining coolant and other contaminants. The proper procedure for doing this is detailed in the sub-section labeled "Steps to Flushing and Refilling the Cooling System" on page 49.

PROBLEMS WITH CONVENTIONAL "MIXED" COOLANTS IN HEAVY-DUTY APPLICATIONS:

Fully formulated coolant eliminates the problems associated with the use of conventional heavy-duty coolants. The first coolants used in heavy-duty engines were very similar to automotive coolants. As heavy-duty engines became more powerful and engineers learned to utilize the energy generated by diesel engines more efficiently, the advancing technology of heavy-duty engines required changes in coolant chemistry. A new heavy-duty engine, or one being serviced, received an "initial fill coolant" that had to be prepared by mixing half low-silicate, heavy duty antifreeze and half purified water to result in a properly pre-mixed engine coolant - but the pre-mixed coolant is not ready for use! The coolant still required a "pre-charge" of a supplemental coolant additive (SCA) added to it. The SCA pre-charge is typically added at 3.0%, by volume. SCA's contains additives not included in the low-silicate antifreeze, such as nitrite, that are essential to trouble-free operation of diesel engines. Furthermore, SCAs that meet ASTM specifications must be capable of protecting a coolant system, at recommended dose, all by themselves. As a result, there is a duplication of some of the inhibitors in the antifreeze. When service technicians continuously add SCAs to a conventional coolant, a saturation point is reached and problems develop.

Problems experienced include:

- Precipitation, or drop out, of inhibitors, especially silicate and phosphate.
- Over concentration of SCA chemicals.
- Premature water pump failures.
- Damaged cylinder liners.
- Damaged radiators.



One of the first and still most common problems that heavy duty technicians observe is "green goo" or, more technically, silicate drop-out. This results from using an automotive (light-duty) antifreeze in heavy-duty engines or excessive addition of supplemental coolant additives. As a coolant continues in service, routine addition of chemicals, called supplemental coolant additives, is necessary. Typically the SCAs are added at 6 month to 1 year intervals. Over concentration sometimes resulted from not changing the coolant while continuing to add SCAs. Premature water pump failure and radiator damage were symptoms of excessive SCAs in the coolant. Frequently, the reverse situation was observed. Operators and technicians often "topped off" coolant systems with plain water or the wrong antifreeze, diluting the SCA concentration. Wet sleeve cylinder liner cavitation and freezing damage resulted from coolant weak in SCAs or glycol.

Fully formulated coolant eliminates these problems because it's fully-formulated technology includes an optimally balanced SCA technology and can be mixed with the drinking water from city water systems. Most of the problems observed with the conventional antifreezes resulted from improper maintenance practices.

- A heavy-duty fully formulated coolant will have a high-quality Supplemental Coolant Additive (SCA).
- This eliminates the need for SCAs at initial fill and when topping off.
- A heavy-duty fully formulated coolant contains nitrite for superior wet sleeve liner cavitation protection.
- A heavy-duty fully formulated coolant will be a phosphate-free formula that reduces the risk of scale and meets Detroit Diesel requirements for a phosphate-free coolant.
- A heavy-duty fully formulated coolants low silicate and low total dissolved solids formulation prevents drop out and gel.
- A heavy-duty fully formulated coolant has the advantage of lower dissolved solids, improved water pump life and complete compatibility with conventional antifreezes/coolants.
- A heavy-duty fully formulated coolant should meet ASTM D-6210, the specification for a fully formulated coolant for use in heavy duty applications.
- A heavy-duty fully formulated coolant should meet TMC RP 329, the specification for a fully formulated coolant which does not require a SCA pre-charge to be added before use in heavy duty engines.

"Low Silicate Heavy Duty Engine Coolant" is defined by ASTM specification D 4985. A common example was the public formulation known as GM 6038-M, which is now obsolete but still commonly referenced. Other mixtures are possible from 33% antifreeze to 67% antifreeze.

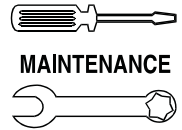
FULLY FORMULATED COOLANT/ANTIFREEZE - WHY?

The new technology - Fully Formulated Antifreeze/Coolant:

Most maintenance managers have different kinds of engines to maintain. These may include cars, pickups, large trucks, and heavy-duty/industrial engines. For years, maintenance managers had to purchase, inventory, and correctly use two or three different types of antifreeze to meet the needs of the different engines in their inventory. Even more difficult, the largest trucks and heavy-duty/industrial engines required a low silicate coolant that needed to be mixed with a separate supplemental coolant additive (SCA) and deionized water. Many industry and professional organizations involved with heavy duty engines and coolants have been discussing the benefits of using a SCA precharged, or fully formulated, antifreeze/coolant. Fully formulated antifreeze, offers convenience, low cost and performance benefits as detailed below.

Background:

To reduce engine downtime and lost revenue, heavy-duty/industrial engines should use a coolant specifically formulated for use in diesel engines. This type of antifreeze is referred to in the industry as a "low silicate" antifreeze, or by ASTM D-4985, the specification numbers that were established to define its characteristics. However, a low-silicate antifreeze, by design, requires the addition of a separate supplemental coolant additive (SCA) before it goes into service. Chemistries of competing antifreezes that meet the ASTM D-4985 specification vary, so SCA's must be engineered to be compatible and complementary to many different antifreezes. As a result, some overlap of chemistry is inevitable in the finished coolant. Furthermore, the blending of antifreeze, water and SCA's is difficult to test and monitor. It is possible for too much or too little of one of the three components to be used, resulting in either inadequate protection or problematic chemical imbalances. Fully formulated coolant solves these problems.



Fully Formulated Coolant:

To eliminate the confusion and difficulties of performing coolant blending on the shop floor, as well as to provide an optimum chemical formulation, fully formulated coolant was developed. Fully formulated coolant meets or exceeds all heavy duty and automotive antifreeze specifications. It is easy to use because it is simply mixed 50/50 with water and poured into the cooling system. It is precharged with the necessary base chemistry and the SCA chemistry needed to provide complete system protection for virtually any engine.

Resulting Benefits:

- No misapplications.
- Simplifies maintenance procedures. One antifreeze for all the engines in the inventory; large and small, gas or diesel.
- Better quality control, no damage to forgetting the SCA pre-charge at initial fill and when topping off.
- Reduced cost compared to mixing components.
- Half the inventory investment.
- Low silicate and low total dissolved solids formula improves water pump and radiator life.
- Phosphate free formula reduces the risk of scale.
- Longer coolant life.
- Less chance of inhibitor instability.
- State of the art corrosion inhibitors eliminate the need for expensive deionized water.
- Complies with engine OEM requirements.



- ⚠ Do not remove the radiator pressure cap while the engine is hot or serious burns from boiling liquid or steam could result.
- ⚠ Ethylene glycol base antifreeze is poisonous. Do not use your mouth to siphon coolant from the radiator, recovery bottle or any container. Wash your hands thoroughly after handling.
- ⚠ Never store used antifreeze in an open container because animals are attracted to the smell and taste of antifreeze even though it is poisonous to them.



- ⚠ Do not use any chromate base rust inhibitor with ethylene glycol base antifreeze, or chromium hydroxide ("green slime") will form and cause overheating. Engines that have been operated with a chromate base rust inhibitor must be chemically cleaned before adding ethylene glycol base antifreeze. Using any high silicate antifreeze boosters or additives also will cause overheating. We also recommend that you DO NOT use any soluble oil inhibitor for this equipment.

STEPS TO FLUSHING AND REFILLING THE COOLING SYSTEM

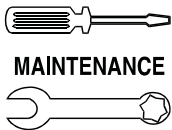
1. Clean the radiator

Make sure the engine is cool before you start.

Your engine/generator's cooling system sucks in air as it operates, which means insects, dirt and leaves can be pulled or pushed into the fins of the radiator. This condition prevents air from passing easily through the radiator, and can cause your engine to overheat.

- Using a hose with a nozzle, a bucket of soapy water and a soft nylon brush, loosen the insects, leaves and other debris. Gently brush with the direction of the fins of the radiator fan, not against it. These thin metal fins are fragile and can be bent with this action if you're not careful.
- Follow this gentle cleaning with a gentle stream of water on the front and rear of the radiator. Don't turn up the hose to full pressure--a concentrated blast of water can actually bend the fins of the radiator.

In general, you should clean the radiator fins every year. Don't do it when the engine is hot, cold water can damage a hot engine. Wait until the engine has cooled before you start cleaning.



2. Center drain pan under the drain spout.

- Never allow coolant to drain out onto the ground. Its sweet smell attracts animals that will drink it. Don't leave it to drain unattended in a pan for the same reason. Another way to protect wildlife from accidental spills is to use antifreeze that utilizes propylene glycol, instead of ethylene glycol. Antifreeze that is made with propylene glycol is safer for pets and people.
- The pan should be big enough to hold the coolant, but not so deep that it doesn't lie flat or slide easily into place. Tipping a pan on its edge (to slip it into position) is a bad idea. It will only lead to spills.

3. Remove the radiator pressure cap

- Cover the radiator cap with a rag. You've made sure the engine is cold, but go slowly anyway.
- With the heel of your hand, press down firmly, turn the radiator cap a quarter turn to the left. You'll feel the cap is loose when you move it, but it's still locked on. (This first turn is to release any pressure in the radiator.)
- Then, pressing down once more, turn the cap another quarter turn to the left and take it off.

Figure 55 — Radiator Cap



Figure 56 — Radiator Cap



4. Inspect the pressure cap, clamps & hoses

The radiator cap acts as more than just a lid for your radiator; it keeps your engine cool by sealing and pressurizing the coolant inside. And not all caps are the same--they are made to maintain a variety of pressures, depending on the size and type of engine. How do you know the pressure rating of your cap? It's written on the top of the cap. When you replace it, make sure you get one that has the same rating.

You will notice that the cap has three main features. The wider top of the cap, the smaller seat at the bottom (with a rubber seal) and a spring coil in between. This spring is what seals the cap to maintain pressure. If you squeeze the seat and the top together, and it's very easy to compress the spring, then replace the cap. It isn't doing its job.

If your cap is rusted or the rubber seal is dried out, it should be replaced.

There are two radiator hoses you'll want to check. The hose at the top of the radiator carries the hot coolant away from the engine, and the hose at the bottom carries the cooled coolant back to the engine.

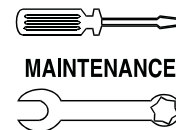
Inspect the radiator hoses for leaks, cracks or soft, mushy condition. Look for signs of rust on the clamps that hold the hoses.

If you find one bad hose, the other hose will probably go bad in the near future. Because the coolant must be drained to replace the hoses, it makes sense to replace both at the same time. But don't actually replace the hoses until you drain the radiator.

5. Drain the radiator

To drain the radiator, you'll need to locate the radiator's drain valve. You'll find it on the bottom of the radiator (some units have remote radiator drains). **Pay attention to how much drains from radiator. It will indicate much you will need to replace later on.**

- Open the drain valve. As you open this valve, the coolant will start to flow from the radiator into the drain pan you placed below it.



- Allow enough time for all of the coolant to drain out, and then close the drain valve.
- Using the funnel, pour the drained coolant into a container that meets the disposal regulations in your area. Set it in a safe place while you continue this project.

Figure 57 — Radiator Drain



Figure 58 — Remote Drain

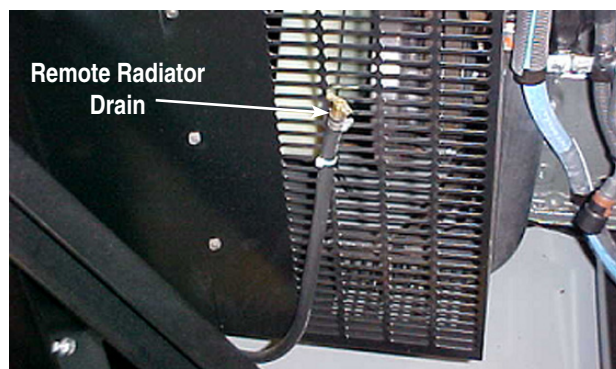
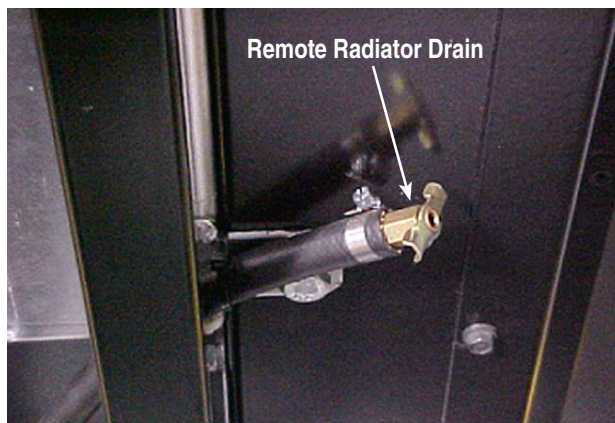


Figure 59 — Remote Drain



6. Rinse/Flush the radiator and cooling system

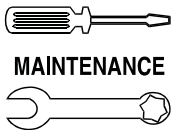
Oddly enough, the actual rinsing of the radiator is the easiest part of the entire operation. Your garden hose does all of the work--you just pay attention to the color of the rinsing water as it leaves the radiator.

- Take the garden hose and insert it into the fill spout.
- With the radiator drain valve closed, fill the radiator until full. Then open the drain valve and drain it once again.
- Important: collect this first rinse and dispose of it in the same manner as the drained coolant. That is, add it to the disposal containers that you have for Step 5.
- If the drained water is clear, close the drain valve and move onto Step 7. If it appears rusty, continue filling and draining until the water is clear. After the first flush, it's alright to let the rinse water drain on to the ground.
- Close the drain valve.

Alternate Step 6: Rinse and replace a hose

Have you removed the bottom hose to replace it? Don't put the new one on until you've rinsed out the radiator. In this case, you want to rinse the radiator with the petcock always closed. (If you're replacing a top hose, go ahead and replace the hose and follow the procedure described above).

- Hold your gloved hand over the hole on the radiator where the bottom hose attaches.
- Fill the radiator with the hose.



- Release your hand quickly. The water will gush out and any rust or mineral deposits will escape through the larger opening of the hose. Repeat this until the water runs clear.
- Replace hoses and clamps as needed. Fortunately this is a simple matter--just unscrew the old clamps on both ends of each hose, and the hoses come right off. And installing the hoses is the same motions in reverse.

7. Add the coolant and water mixture

The optimum coolant combination is a 50/50 mixture of antifreeze and water. The 50/50 mixture will produce freeze protection down to -34F (-36C), and boil over protection up 265F (129C). However, in colder climates, where lower temperature freeze protection is needed, a mixture of up to 70% antifreeze can be used.

Although regular water will do, the refilling of the system should be done with distilled water. Distilled water doesn't contain any minerals which can dissolve and cause scale and deposits in your cooling system.

There are four ways (at least) to refill the radiator:

- Fill the radiator with a fully formulated pre-mix antifreeze. This is by far the easiest way to fill the radiator and insure that the system is filled with a 50/50 mixture of antifreeze and water.

A second way:

- Fill the radiator halfway with water. Can't see into the radiator? Estimate the amount of water based on the amount of coolant that you drained out. If you drained out two gallons (8 liters) of coolant, then you need to add one gallon (4 liters) each of water and antifreeze.
- Using a funnel, fill up the radiator with antifreeze.

A third way:

- In a separate clean container, mix antifreeze and water in a 50/50 ratio.
- Add the 50/50 mixture until the radiator is full, and store any extra coolant.

A fourth way:

- Pour in a half container of antifreeze. This usually comes in one-gallon (4 liter) containers, so add a half-gallon (2 liters).
- Add to the radiator an equal amount of a distilled water. Then mix water into the antifreeze container (which still has antifreeze in it). This will create the 50/50 mix, which you can add to the radiator until it's full.

It's good idea to have a little extra coolant after this fill-up, because the coolant level will drop slightly after you do Step 8. The excess from this step can be used to top off the radiator a little later.

8. Bleed the system

It's likely that pockets of air will be in the cooling system after you've refilled the radiator. The system is self-bleeding when you follow these steps:

- Start your engine, but leave the radiator cap off so the pressure doesn't build up.
- Let the engine run until it reaches its operating temperature of about 190F (88C). This'll take about 15 minutes.
- Check the level of the coolant. You may find that the level has gone down some. How did that happen? The air trapped in the radiator kept the level artificially high; when this air was released, it opened up more space in the radiator, and allowed more antifreeze into the system. Add more water and antifreeze, in the same 50/50 ratio, to fill the space left by the bled-out air.
- As the air bleeds out, the coolant may bubble and spit from the opening on top of the radiator. Use caution--that coolant is hot.

Allow the engine to cool. Then check the freeze/boil over protection with an antifreeze tester. Add more antifreeze or water to the system as necessary. Don't forget to clean out and refill your coolant reservoir with a 50/50 mixture of antifreeze and water as well.

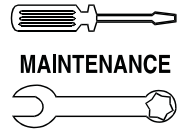


Figure 60 — Coolant Recovery Tank



9. Replace the pressure cap

Basically, this isn't too tough--you replace the cap in reverse of how you removed it. It's important, however, that you get it on tightly. The cap has two wide tabs, opposite each other on the cap, hanging down slightly. Measured across, each tab is about one quarter of the circumference of the cap. The fill spout, where you poured in the coolant, has two notches or cutouts that match up with the tabs of the cap.

- Match the tabs with the cutouts on the fill spout. Press down firmly on the cap and turn it to the right a quarter turn. Give it a jiggle to be sure it's in place. Though the cap shouldn't be tight at this point, you shouldn't be able to lift it off, either.
- Turn the cap another quarter turn to the right, still pressing down, until you feel it lock into place. It should be locked tight now and won't move if you try to wiggle it.

10. Clean up

- Check for spills of antifreeze, cleaner or coolant mixture. Use the hose to dilute any spills that might have inadvertently occurred.
- Dispose of old clamps and hoses.
- All rules and regulations should be observed when disposing of antifreeze/coolant and radiator fluids. Call the EPA or the office of your state or local environmental agency for details on disposal procedures. Until then, put the containers up out of the reach of children and animals.

Congratulations, you've just protected your engines cooling system for at least the next two years!

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